MIL-T-81527B(AS)
29 January 1971
SUPERSEDING
MIL-T-81527A(AS)
4 December 1970

MILITARY SPECIFICATION

TEST SET, MISSILE CONTROL SYSTEM, AWG - 10 SIDE BENCH

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. The equipment listed herein shall be an integrated assembly of Contractor - Furnished and Government-Furnished Equipment to facilitate detailed maintenance of AN/AWG-10 Missile Control System (MCS) at the Intermediate Level where a sidebench is available. Furthermore, an AN/AWG-10 MCS Test Set, of and by itself, does not fulfill complete maintenance requirements for the Intermediate Level Maintenance of the AN/AWG-10 MCS. Facilities having a centerline bench should refer to MIL-T-81517B(AS), Test Set, Missile Control System AN/AWM-21.

1, 2 Classification. The AWG - 10 Side Bench Missile Control System Test Set shall consist of the following items:

ITEMS	TYPE DESIGNATION
Table -Rack, Radar Maintenance	FN-212E/AWG-10
Fixture, Antenna Maintenance	MX-7340C/AWM-21
Wave guide Extension	MX-8491/AWG-10
Cable and Adapter Group	OA-8124B/AWM-21
Tool Kit, Test Set	TK-217/AWM-21
Toot Kit, Maintenance	TK-216/AWM-21
Fixture, Antenna Transporting	MX-7339B/AWM-21
Test Set, Reference Rate Gyro	TS-2512/AWM-21
Generator, Signal	SG-6y6B/AWM-21
Compressor-Dehydrator, Electronic Equipment	HD-755/AWM-21
Test Set, Missile Control System (MCS)	TS-2511C/AWG-10*
Counter, Electronic Digital	CP-870/AWM-21
0scilloscope	OS-180/AWM-21
Analyzer, Radio Frequency Noise	TS-2573E/AWG-10
Cabinet, Electrical Equipment	CY-6280B/AWM-21*

*Pending Navy Approval

FSC-4935

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ITEMS	TYPE DESIGNATION
Generator, Signal	SG-844/AWM-21
Voltmeter, Digital	ME-371/AWM-21
Radar Set	AN/APG-59
Multimeter	AN/PSM-4
Multimeter	TS-505/U

1. 2.1 Government Furnished Equipment. The Radar Set AN/A PG -59, Multimeter AN/PSM-4 and Multimeter TS-505/U shall be Government Furnished Equipment (GFE).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-17 - Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead.

MIL-P-642 - plugs, Telephone.

MIL-C-3767 - Connectors, Electrical (Power, Bladed Type) General Specification for.

MIL-T -5422 - Testing, Environmental, Aircraft Electronic Equipment.

MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and ordinances

MIL-I-6181 - Interference Control Requirements, Aircraft Equipment.

MIL-H-15362 - Harness; Test Bench, Avionic Equipment, General Specification for.

MIL-W-16878 Wire, Electrical, Insulated, High Temperature.

MIL-M-17096 Multimeter, AN/PSM-4.

MIL-E-17555 Electronic and Electrical Equipment and Associated Repair Parts, Preparation for delivery of.

MIL-T-18303 Test Procedures: Preproduction and Acceptance for Aircraft Electronic Equipment, Format For.

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MIL-T-21200 -Test Equipment for Use with Electronic and Fire Control Systems, General Specification for.

MIL-C-22750 - Covering Epoxy Polyamide

MIL-S-22885 -

MIL-T-23336 - Technical Directives (Letter Type): Preparation Of.

MIL-C-24165 - Counter, Electronic, 500 Megacycle, AN/USM-207

MIL-S-52059 -Shelter, Electrical Equipment S-141( )/G

MIL-M-55088 - Multimeter, Electronic TS-505( )/U

MIL-A-81030 - Air Transportable, Shelter Vans, For Dressing and briefing, Dental and Sick Bay, Aircraft Maintenance, Serological.

MIL-R-81114B - Radar Set AN/APG-59.

MIL-T-81516 Test Set, Missile Control System AN/AWM-22

MIL-T-81517 - Test Set, Missile Control System AN/AWM-21

MIL-C-81583 - Cabinet, Electronic Equipment

MIL-T-81741 - Test Set, Missile Control Group OQ-50/AWM-21, OQ-51/AWM-21, and OQ-52/AWM-21

STANDARDS

Military

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

MIL-STD- 189 - Racks, Electrical Equipment, 19-inch and Associated Panels.

MIL-STD-781 - Reliability Test, Exponential Distribution.

MIL-STD-803 - Human Engineering Design Criteria for Aerospace Systems and Equipment.

Federal

FED-STD-595 - Colors

DRAWINGS

Military

MS-3100 - Connector Electrical

MS-3102 - Connector, Receptacle, Electric, Box Mounting

MS-3126 - Connector, Plug, Electric

MS-17322 - Meter, Time Totalizing

MS-25307 - Switch, Toggle, Positive Break Aircraft, Double Pole.

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MS-33656 - Fitting End, Standard Dimension for Flared Tube.

NAMC 610415 - Shipboard Workbench for Aviation Electronics Shops

2.1.1 Availability of Documents - Copies of Specifications, Standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office (see 6. 4).

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. in so doing the following shall govern:

(a) Parts and Materials as approved by MIL-T-21200 shall be given first consideration.

(b) Guidelines for the selection of nonstandard parts shall be per MIL-T-21200, Selection of parts and materials.

(c) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass electrical and' environmental test.

3.1.1 Nonstandard Parts - Nonstandard parts and materials shall be as outlined in MIL-T-21200 Materials, parts and processes, except that mechanical parts shall be considered as nonstandard parts only when the material requirements of MIL-T-21200, as modified herein, are not met.

3.1.2 Tube Types and semiconductors Types - The choice and application of electron tubes and semiconductor types used in the equipment shall be in accordance with MIL-T-21200 and MIL-STD-701.

3.1.3 Noncorrosive Hardware - External, Military Standard (MS) hardware which is used during servicing or installing test equipments shall be of a corrosive-resistance type and shall not be further finish-treated.

3.1.4 Paint - Paint shall be epoxy polyamide per MIL-C-22750; color shall be aircraft gray, color number 16473, per FED-STD-595.

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3.2 Design and Construction - Unless otherwise specified herein or in the detail equipment specification, Contractor-Furnished Equipment (CFE) shall conform with all the requirements of MIL-H- 15362 or MIL-T-21200 as applicable for design, construction, and workmanship. Government - Furnished Equipment (GFE) shall conform to the applicable detail equipment specification. The total weight of the AWG-10 Side Bench Missile Control System Test Set shall not exceed 5, 000 pounds.

3.2.1 Operational Stability - Unless otherwise specified in the individual equipment specification, the equipment shall operate with satisfactory per - formance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use. However, adjustments at points inaccessible to the operator during use may be made following replacement of parts.

3. 2.2 Operating Life - The equipment shall have minimum total operating life of 2,000 hours with reasonable servicing and replacement of parts,

3. 2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-H-15362 and MIL-T-21200, as applicable.

3. 2.4 Interchangeability - Unless otherwise specified herein or in the detail equipment specification, the equipment shall meet the interchangeabil- ity requirements as defined in MIL-T-21200.

3. 2.5 Maintenance provisions and Field Testing - Provisions for maint- enance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. The color of front panel indicator lights shall be in accordance with MIL-STD-803.

3. 2.6 Calibration - Test and Measuring equipment shall be calibrated at periodic intervals established on the basis of stability, purpose, and degree of usage. Calibration intervals shall be adjusted as required to assure continued accuracy as evidenced by previous calibration results.

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3.2.7 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient: 25° +15° -5° C 77° +27°, -9° F
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90 percent relative humidity
Input Voltage	115 ± 1 volts (V) line to neutral, 3-phase, 400 Hz ± 2 percent 27.5 ± 0.5 volts direct current (Vdc)

3.2.8 Warmup Time. Unless otherwise specified herein or in the detail equipment specification, the time required for the equipment to warmup prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions, 20 minutes at extreme service conditions.

3.2.9 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of MIL-I-6181.

3.3 Performance.

3.3.1 General. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein or in the detail equipment specification.

3.3.2 Function. The AWG-10 Side Bench Missile Control System Test Set shall provide an integrated maintenance facility. This shall include

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simulated aircraft controls and equipment test bench harnesses; equipment support and handling fixtures; test equipment both special and standard; special tools; cooling equipment; and a hydraulic power supply.

3. 3.3 Form Factor. The AWG-10 Side Bench Missile Control System Test Set shall be designed for installation in a van as described in MIL-S-52059B, as well as ship and shore avionics shop layouts. Guidance Plan CVA61 Avionics Shop No. 1, Main Deck, FRS 29-39, Drawing 610415, Bench Space 119 through 126 shall be used as a guide to shop layout and bench design.

3.4 Detail Requirements.

3.4.1 Table-Rack, Radar Maintenance.

3.4.1. 1 Function. The Radar Maintenance Table -Rack shall be for shop use and shall provide monitoring, mounting, inter connection, hydraulic power and cooling facilities for the active Line Replaceable Units of the MCS mounted on the bench. The equipment shall electrically simulate MCS required air craft flight functions and shall enable fault is elation, adjustment, and repair or replacement of defective pallets, circuits, plug in assemblies, and components of the MCS.

3.4.1.2 Form Factor. The Table -Rack shall be compatible with the space requirements of Type II Shelters (Aircraft Maintenance) per MIL-A-81030 and shall require a space of 18-1/2 feet by 7-1/2 feet with 6-1/2 feet height. The operating and maintenance area will be 21- 1/2 feet by 10 feet. The total weight of the equipment, including cables, shall not exceed 4, 200 pounds.

3.4.1.3 Mean Time Between Failures (MTBF). When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate at least 400 hours MTBF. Accept-reject criteria shall be defined by paragraph 4. 2.8.1 of MIL-STD-781.

3.4.1.4 Service Conditions, The Equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H- 15362 except that the panels of the Test and Control Rack shall comply with MIL-T-21200, Class 3.

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3.4.1.5 Primary Input power Requirements. The equipment shall require power of the following type and shall not exceed the specified amounts.

AC Power: 3-phase, 115/200V ± 5 percent, 380 to 420 Hz, 85 Amperes (A)/Phase

DC Power: 27 ± 2 Vdc, 70 A

3.4.1.6 Contents of Unit. The Table -Rack shall consist of the following major units:

(a) Support Assembly, RF Section
(b) Bench Assembly, Pallet Section
(c) Test and Contro1 Rack
(d) Indicator Bench
(e) Computer Bench
(f) Hydraulic Supply
(g) Liquid Cooling System
(h) Test Accessories
(i) Antenna Support Fixture
(j) Interconnecting Test Harness
(k) System Harness

3.4.1.6.1 Support Assembly, RF Section.

3.4.1.6.1.1 Function. The RF Section Support Assembly shall mount the RF Section of the AN/APG-59 Radar Set and shall make it accessible for use in the fault isolation, adjustment and repair or replacement of defective circuits, plug-in assemblies, and components of the RF Transmitter Assembly.

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3.4.1.6.1.2 Form Factor. The RF Section Support Assembly shall be in the form of a column and a frame. The frame shall be attached to the column by hinges which shall permit rotation of the frame in the vertical axis to allow free access to the sides and rear of the RF Section of the Radar Set. The column shall be set on a base not to exceed 12 inches by 13 inches and shall be capable of being bolted or welded to the deck. A bracket shall be installed at the top of the column to secure it to a bulkhead or equivalent structure. Two brackets, each not to exceed 7- 1/2 inches by 4 inches by 10-5/8 inches, designed to be attached to a bulkhead or equivalent structure, shall be furnished as part of the Support Assembly. These brackets shall be used as bearing surface for the free end of the frame, when stowed. The frame, when stowed, shall be secured to the two brackets by means of quick release pins. The overall dimensions of the Support Assembly, including column, frame brackets, but without the RF Section of the Radar Set attached, shall not exceed 42 inches by 13-11/16 inches by 71 -1/2 inches when secured against the bulkhead. A clevis shall be installed on the bottom of the frame to permit the frame to be secured at an angle of 90 degrees from the plane of the bulkhead. The mating parts of the clevis shall be attached to the Pallet Section Bench Assembly (see section 3.4.1.6. 2). Cooling air for the RF Section of the AN/APG-59 Radar Set shall be supplied from the Blower Assembly (see section 3.4.1.6.2. 8) by means of a hose connection inlet at the rear of the frame, and distributed through the bottom horizontal member of the frame structure used as a manifold to supply outlets on the front of the frame. The supply outlets on the frame shall be opposite the air inlets of the subassemblies of the Radar Set RF Section. The weight of the Support Assembly shall not exceed 100 pounds, excluding the Radar Set RF Section.

3.4.1.6.2 Bench Assembly, Pallet Section

3.4.1.6.2.1 Function. The Bench Assembly, Pallet Section shall mount the sever al pallets of the System Assembly and the Low Voltage Power Supply of the AN/A PG-59 Radar Set and shall make them accessible for fault

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isolation, adjustment and repair or replacement of defective circuits, plugin assemblies, and components. The Bench Assembly shall provide suitable work space to accomplish these tasks. The Bench Assembly shall mount support equipment including the Blower Assembly and the Remote Safety Unit and shall provide an alternate location for mounting the Test Set, MCS TS-2511C/AWG-10.

3.4.1.6.2.2 Form Factor. The Bench Assembly, Pallet Section, shall not exceed 53-1/8 inches by 20 -1/2 inches by 75 inches overall with the Drop Leaf folded. The weight of the Bench Assembly shall not exceed 500 pounds.

3.4.1.6.2.3 Contents. The Bench Assembly, Pallet Section, shall consist of the following items :

(a) Base Structure
(b) Rear Frame
(c) Pallet Shaft Assembly
(d) Blower Assembly
(e) Remote Safety Unit
(f) Test point Bracket

3.4.1.6.2.4 Base Structure.

3.4.1.6.2.4.1 Function. The Base Structure shall serve as the basic mechanical support for the Pallet Section Bench Assembly, and shall provide a work area, an alternate mounting for the Test Set, MCS, TS2511B/AWG-10, and attachment points for the Rear Frame and the Pallet Shaft Assembly. Space shall be provided below the Base Structure for the Blower Assembly.

3.4.1.6.2.4.2 Form Factor. The Base Structure shall be capable of being either bolted or welded to the deck. The basic structure shall not exceed

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32 inches by 25 inches by 51 inches. The structure shall contain a cutout at the right rear to provide space for the installation of the column of the RF Section Support Assembly. A clevis and quick release pin shall be installed on the structure. It shall mate with the clevis on the Support Assembly to secure the frame of the RF Section Support Assembly in the 90-degree extended position. Attachment points shall be provided at the rear of the Base Structure for the Rear Frame. The Base Structure shall be designed to mount the Pallet Shaft Assembly and to secure it in various service positions.

3.4.1.6.2.4.3 Contents. The Base Structure shall contain the following substructures:

(a) Bench Top

b) Drop Leaf

(c) Cradle Slide Assembly

(d) Power Supply Support

(e) Grounding Wand

3.4.1.6.2.4.3.1 Bench Top.

3.4.1.6.2.4.3.1.1 Function. The Bench Top shall form the basic work surface of the Receiver Section Bench Assembly.

3.4.1.6.2.4.3.1.2 Form Factor. The Bench Top shall be a nominal 51 inches by 28 inches and its surface shall be a nominal 32 inches above the deck. The surface of the Bench Top shall be constructed of masonite bonded to aluminum. The configuration of the Bench Top shall conform to the outline of the Base Structure including the cutout section reserved for the RF Section Support Assembly. The Bench Top shall have a filler section to allow the Pallet Shaft Assembly to be mounted on the Bench Top.

3.4. 1.6.2.4.3.2 Drop Leaf.

3.4. 1.6.2.4.3.2.1 Function. The Drop Leaf shall expand and enlarge the work surface area of the Receiver Section Bench Assembly and shall serve

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as a work surface for various removable assemblies of the Test and Control Rack.

3.4.1.6.2.4.3.2.2 Form Factor. The Drop Leaf shall be installed on the forward left edge of the Base Structure. It shall provide nominal surface area of 29 inches by 12 inches when extended, level with the Bench Top. When folded, the Drop Leaf shall not protrude more than 1-1/4 inches from the front of the Base Structure. Latches and locking devices shall be installed to secure the Drop Leaf in either the extended or the folded position. The surface of the Drop Leaf shall be fabricated from the same material as the Bench Top.

3.4.1.6.2.4.3.3 Cradle Slide Assembly,

3.4.1.6.2.4.3.3.1 Function. The Cradle Slide Assembly shall support the Test Set MCS TS2511B/AWG-10 below the right front corner of the Bench Top. This location will permit the use of the Test Set MGS TS2511C/AWG-10 in the performance of tests associated with the AN/APG-59 Radar Set Assembly Line Replaceable Units (LRU) installed on the Pallet Shaft Assembly and on the RF Section Support Assembly. The Cradle Slide Assembly shall secure the Test Set MCS TS251lC/AWG-10 with its front panel upward and controls easily accessible.

3.4.1.6.2.4.3.3.2 Form Factor. The Cradle Slide Assembly shall not exceed 24 inches by 14 inches by 9 inches when stowed. The Assembly shall consist of two slide rails and a cradle. Latches shall be provided to secure the cradle in the stowed position. The cradle shall pull out a nominal 12-7/8 inches from the front of the Base Structure for use in the extended position.

3.4.1.6.2.4.3.4 Unregulated Power Supply Support.

3.4.1.6.2.4.3.4.1 Function. The Unregulated Power Supply Support shall support the Unregulated Power Supply, PP-455l/APG-59 in the horizontal position above the deck. The Support shall be at the right side of the Base Structure beneath the Cradle Slide Assembly.

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3.4. 1. 6.2.4.3.4.2 Form Factor. The Power Supply Support shall consist of two angle brackets a nominal 15- 1/4 inches long attached to the Base Structure. They shall support the power supply at a nominal 7-1/2 and 8 inches above the deck, front and rear respectively.

3.4. 1. 6.2.4.3.5 Grounding Wand. A high-voltage Grounding Wand 19 ± 0.5 inches long shall be mounted on the Base Structure. It shall be electrically connected to the main ground bus of the bench and shall permit grounding of all high voltage terminals of the RF Section of the AN/APG-59 Radar Set.

3.4. 1. 6.2.5 Rear Frame.

3.4.1.6.2.5.1 Function. The Rear Frame shall form the rear and overhead structure of the Receiver Section Bench Assembly. It shall provide the upper attachment points for the Pallet Shaft Assembly, attachment points for the Convenience Power outlet Installation, the Remote Safety Unit and a test point bracket and brackets and clamps for the interface between the Inter-connecting Test Harness and the System Harness.

3.4. 1. 6.2.5.2 Form Factor. The Rear Frame shall be attached to the rear of the Base Structure with its bottom edge a nominal 4 inches above the deck. The rear vertical portion of the frame shall be nominal 39 inches by 68 inches less attachments. The overhead structure shall not exceed 39 inches by 24- 1/2 inches by 3 inches. To improve access to the pallets, the sides of the overhead structure shall be cut back 12-7/8 inches. Both the vertical portion and overhead structure of the Rear Frame shall form an integral structure. The overhead structure shall support the upper end of the Pallet Shaft Assembly. Latches shall be provided to secure the pallets of the AN/APG-59 Radar Set in the work position. An attachment point shall be provided for the Remote Safety Switch on the structural member, which defines the right cut-back portion of the overhead structure. The attachment point shall be provided for a test point bracket on the left portion of this same member.

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3.4. 1.6.2.5.3 Contents. The Rear Frame shall contain the following substructures:

(a) Convenience Power Outlet Installation

(b) Interconnecting Test Harriess Shelf

3.4.1.6.2.5.3.1 Convenience Power Outlet installation.

3.4.1.6.2.5.3.1.1 Function. The Convenience Power Outlet installation shall make utility power available at the Pallet Section Bench Assembly, and shall provide the connection point to the user' s source of supply. The Convenience Power Outlet Installation shall also serve the RF Section Support Assembly.

3.4.1.6.2.5.3.1.2 Form Factor. The Convenience Power Outlet Installation shall consist of receptacle strip installed in a crossmember of the Rear Frame 37 inches above deck level and a terminal strip with 3 terminal points installed on the upper left side of the Rear Frame. The receptacle strip shall contain eight convenience power outlets rated at 115 VAC, 25A, The outlets shall be 2-pole, 3-wire, polarized and shall meet the requirements of MIL-C-3767. These outlets shall receive power directly from the primary ac power source and shall not be connected through the circuit breakers but will be separately connected by the user.

3.4.1.6.2.5.3.2 Interconnecting Test Harness Shelf.

3.4.1.6.2.5.3.2. Function. The Interconnecting Test Harness Shelf shall support portions of the Interconnecting Test Harness and the System Harness which lie in the vicinity of the Receiver Section Bench Assembly.

3. 4.1.6.2.5.3.2,2 Form Factor. The Interconnecting Test Harness Shelf shall consist of two panels atop the overhead structure of the Rear Frame with a bracket located atop the Rear Frame. This bracket shall accommodate and support the connectors which form the interface between the Interconnecting Test Harriess and the System Harness. Each of the panels

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shall not exceed 16 inches by 24 inches and shall conform to the shape of the Rear Frame overhead structure.

3.4.1.6.2.6 Pallet Shaft Assembly.

3.4.1.6.2.6. 1 Function. The Pallet Shaft Assembly shall serve to support the three pallets of the Receiver Section of the AN/APG-59 Radar Set Barrel Assembly for service and adjustment. The Pallet Shaft Assembly shall also distribute cooling air to these pallets through a central shaft and associated manifolds.

3.4.1.6.2.6.2 Form Factor. The Pallet Shaft Assembly shall consist of a rotable hollow shaft, nominal 4-3/4 inches in diameter by 44 inches high. The shaft shall rest on load-bearing surface built into the Base Structure below the Bench Top and shall be supported by overhead structure of the Rear Frame. The shaft shall contain fittings to support three pallets of the AN/APG-59 Radar Set Barrel Assembly. The three pallets shall be mountable in normal vertical position on bracket structures 90 degrees apart about the shaft. The fourth side of the shaft assembly shall be left open. Each Pallet shall be attached to the bracket structure by means of vertical hinges to permit access to the rear of the pallets for servicing and adjustment. Latches shall be provided for each pallet location to lock the pallets in place to the bracket structures when access to the rear of the pallets is not required. Means to lock the Pallet Shaft Assembly into place shall be provided on the Rear Frame overhead structure. Means shall be provided to lock the pallets in the open position for stability during servicing by means of the lower and upper stow brackets installed on the Base Structure and on the Rear Frame overhead structure respectively.

3.4.1.6.2.6.3 Air Distributions The Pallet Shaft Assembly shall be used as a manifold to distribute cooling air to the pallets. The lower end of the shaft shall be open on the bottom and shall be connected through a rotating air duct to the plenum chamber of the Blower Assembly. The shaft shall be closed at the top to prevent escape of cooling air. The lower bracket

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structure shalI form an air distribution manifold to receive air from the shaft and distribute it to the pallets in all servicing positions, through rotating joints and flexible hoses.

3.4.1.6.2.7 Blower Assembly.

3.4.1.6.2.7. 1 Function. The Blower Assembly shall provide filtered cooling air to the AN/APG-59 Radar Set LRU's when installed on the RF Section Support Assembly and the Pallet Support Shaft Assembly, and to the Unreg - ulated Power Supply, PP-4551/APG-59, when mounted on the Base Structure.

3.4.1.6.2.7.2 Form Factor. The Blower Assembly shall not exceed 28 inches by 20 inches by 25 inches, exclusive of ducting, and shall be mounted on the deck beneath the Base Structure. The weight of the assembly shall not exceed 300 pounds.

3.4.1.6.2.7.3 Service Conditions. The Blower Assembly shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in section 3.4.1.4.

3.4.1.6.2.7.4 Primary Input Power Requirements, The Blower Assembly shall operate from input power of 3-phase, 115/200 V, 400-Hz, 2250 VA maximum.

3.4.1.6.2.7.5 Noise Levels, Acoustical noise levels shall not be produced from this assembly in excess of 85 decibels (dB) (O. 0002 microbar reference) when measured 6 feet from the unit and 5 feet above the deck.

3.4.1.6.2.7.6 Performance. The Blower Assembly shall provide a minimum of 10. 5-pounds per minute (ppm) of air at a pressure head of 12.8 inches of water. The thermal rise across the unit shall not exceed 5.5°C (10° F), and the energy dissipated in the motor shall not be carried into the output air stream.

3.4.1.6.2.7.7 Contents of Unit. The Blower Assembly shall consist of:

(a) Centrifugal Fan

(b) Transition Unit

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(c) Distribution Ducts

(d) Plenum Chamber

3.4.1.6.2.7.7.1 Centrifugal Fan. The Fan shall be designed to operate continuously and shall deliver sufficient air to comply with performance requirement specified in 3.4.1.6.2.7.6. The unit shall be 16-1/2 inches by 25-5/6 inches by 20 inches including the air inlet and exhause unit. The air inlet and exhaust shall be a nominal 4-1/2 inches in diameter. Weight shall not exceed 275 pounds.

3.4.1.6.2.7.7.2 Transition Unit. The Transition Unit shall contain an acoustical baffle and air filter. This unit shall mate the air inlet of the fan through a 9-1/2 inch long duct to an 18-inch by 12-inch air inlet. The air acoustical foam baffle shall reduce the fan acoustical noise to or below the level specified in 3.4. 1.6.2.7, 5.

3.4.1.6.2.7.7.3 Distribution Ducts. The air exhaust of the Centrifugal Fan shall mate with a series of ducts which shall divide the air supply into two parts. One part shall supply the RF Section Support Assembly and the other part shall supply the Plenum Chamber.

3.4. 1. 6.2.7.7.4 Plenum Chamber. The Plenum Chamber shall be installed below the Bench Top directly underneath the Pallet Shaft Assembly. The Chamber shall not exceed 12-1/4 inches by 6 inches by 5 inches and shall distribute air from the Blower Assembly to the AN/APG-59, Radar Set Unregulated Power Supply, PP455l/APG-59, and to the Pallet Shaft Assembly through an appropriate duct. A differential pressure switch shall mount to one end of the Plenum Chamber to sense static pressure differential between the Plenum Chamber and outside ambient air. The switch shall activate when the air pressure decreases to 5.50 + O. 25, -0.00 inches of water and shall be electrically connected to the CMPTR AIR, indicator in series with a sensing element installed in the Plenum chamber of the Computer Support Structure.

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3.4.1.6.2.7.8 Controls. The Blower Assembly shall be activated from the Test and Control Rack by Meither placing the RSC FUNCTION switch to STBY or OPR or by depressing the AIR COOL pushbutton switch.

3.4.1. 6.2.7.9 Connections. The electrical connector shall be MS310OA18-11P,

3.4.1.6.2.8 Remote Safetv Unit,

3.4.1.6.2.8.1 Function. The Remote Safety Unit shall provide the means for an emergency cut-off of power to the system and shall be installed on the front edge of the overhead structure of the Rear Frame to the right of the top support of the Pallet Shaft Assembly. It shall be accessible to operat - ing positions at the pallet bench, the Radio Frequency Noise Analyzer and the RF Section Support Assembly.

3.4.1.6.2.8.2 Form Factor. This unit shall not exceed 5-1/2 inches by 4 inches by 2 inches. The weight of the unit shall not exceed 2 pounds.

3.4.1.6.2.8.3 Controls. The Remote Safety Unit shall provide a red momentary pushbutton switch. When depressed, the switch shall operate in conjunction with the Power Control Panel to remove all power from the Radar Set. When the control has been operated, power can be reapplied only by operation of the Power On Switch (PWR ON).

3,4. 1. 6.2.8.4 Indication. The unit shall have a red warning indication light which shall be continuously lit whenever transmitter high voltage is present in the AN/APG-59 Radar Set. The warning light shall start flash - ing any time conditions are such as to permit RF radiation from the antenna,

3. 4.1.6.2.9 Test Point Bracket.

3.4.1.6.2.9.1 Function. A test point bracket with test points capable of continous monitoring, and associated with the transmitter, shall be provided to back up BIT and FIM comparators.

3.4.1.6.2.9.2 Form Factor. The dimensions of the test point bracket shall not exceed 6 inches by 6.5 inches by 2. 7 inches, and it shall be contained within the nominal dimensions of the Bench Assembly. The weight of the bracket shall not exceed 1 pound.

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3.4.1.6.2.9.3 Test Points. The following test points, BNC-type female connectors, appropriately wired to the respective signals shall be contained on the bracket.

(a) PD Video

(b) PD Sync

(c) Pulse Video (B-Gun Video)

(d) Pulse Sync (TO)

(e) Line Frequency (400 Hz)

(f) Pretrigger Pulse

3.4.1.6.3 Test and Control Rack.

3.4.1.6.3.1 Function. The Test and Control Rack shall provide in one cabinet the necessary controls for primary power, hydraulics, liquid coolant system, and other bench functions; provide master MCS control, air craft simulation and safety interlocks. It shall also contain test equipment mounting provisions.

3.4.1.6.3.2 Form Factor. This shall be a standard 19-inch equipment rack containing the items specified herein. The maximum dimensions of this unit shall be 28- 1/2 inches by 30 inches by 75 inches. The weight of the Test and Control Rack shall not exceed 700 pounds, including the Interconnecting Cabling, Shock Mounts, and Bottom Frame.

3.4.1.6.3.3 Contents of Unit. The Test and Contro1 Rack shall contain the following assemblies:

(a) Rack Assembly

b) Power Contro1 Panel

(c) Aircraft Simulator Panel

(d) MCS Contro1 Panel

(e) Manual BIT Panel

(f) Umbilical Plug and Tuning Drive Test Panel

(g) Voltage Regulator

(h) IFF Simulator Panel

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3.4.1.6.3.3.1 Rack Assembly.

3.4.1.6.3.3.1.1 Function. The Rack Assembly shall provide mounting provisions for the panels listed in contents of unit (see section 3.4.1.6.3.3). It shall provide certain Convenience Power Outlets (see section 3.4.1.6.3.3.1.5).

3.4.1.6.3.3.1.2 Form Factor. The rack frame shall be a standard 19-inch rack per MIL-STD-189. The weight of the entire unit shall not exceed 425 pounds . The vertical shock mounts shall be located between the rack and the Base Frame of the Assembly. The horizontal shock mounts shall be located between the rack and bulkhead or equivalent support structure behind the rack. These mounts shall permit the Test and Control Rack to pass the shock and vibration requirements of MIL-H-15362.

3.4.1.6.3.3.1.3 Cables. The interconnecting cables to the Radar Set Control (RSC), the Selector -Test Programmer, and the Antenna Hand Control which are mounted on the MCS Control Panel, shall be 72 ± 6 inches long. The first two units, while still electrically connected, can be separately removed from their mountings and be laid on the adjacent Drop Leaf of the Receiver Section Bench Assembly.

3.4.1.6.3.3.1.4 Harness. Electrical Inter connections within the Test and Control Rack Shall be accomplished by a harriess.

3.4.1.6.3.3.1.5 Convenience Power Outlets, Two convenience Power Outlet circuits shall be provided on the Rack Structure. Single-phase 115v, 400 -Hz shall be provided by two Convenience Power Outlets. One shall be on the left side and the other on the right side of the Rack Structure. Three -phase 115/200V, 400-Hz shall be provided by one outlet on the right side of the Rack Structure. The outlets shall be connected directly to the primary ac power source and shall not be connected through the circuit breakers.

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3.4.1.6.3.3.1.6 Connectors. Connections to the AN/APG-59 Radar Set assemblies mounted an the MCS Control Panel shall be made through the following connectors:

Reference Designation	Function
12P1*	Radar Set Contro1
12P2* 6&8	Radar Set Control
12P3* 2,6,&8	Radar Set Control
13 P1*[6]	Antenna Contro1
14P1	Selector-Test Programmer
14P2	Selector-Test Programmer

3.4.1.6.3.3. 1.7 Shipping List, A list, of a durable nature, shall be affixed to the side of the Test and Control Rack, The list shall be comprised of the major assemblies and the numerous small accessories which comprise a complete Table Rack Radar Maintenance. The list shall be for inventory control, especially whenever the bench is transported or shipped.

3.4.1.6.3.3.2 Power Control Panel.

3.4.1.6.3.3.2.1 Function. This panel "shall provide the power control and monitoring for the Table -Rack.

3.4.1.6.3.3.2.2 Form Factor. This unit shall be 19 inches by 14 inches by 7 inches. The weight of the unit shall not exceed 27 pounds.

3.4.1.6.3.3.2.3 Controls. The controls listed below shall be provided on the front panel. System response to the operation of the controls listed in (a) through (h) below shall be indicated by the illumination of the control itself.

(a) PWR ON (Power On) - A green pushbutton switch shall transfer +28 Vdc to power control relays thus enabling power to be supplied through-out out Table-Rack.

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*2 See paragraph 6. 5.2
6 See paragraph 6. 5.6
8 See paragraph 6. 5.8

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(b) PWR OFF (Power Off) - A red pushbutton switch shall remove +28 Vdc from the Power Control Relays thus enabling all power to be removed from the Table -Rack.

(c) RDR PWR (Radar Power) - The Radar Power Control shall be a green pushbutton switch which shall be used to control the 115/200v, 3-phase 400 Hz and the 28 Vdc power to the MCS.

(d) PNL LTS (Panel Lights) - The Panel Lights Control shall be a white pushbutton switch which shall be used to turn the MCS cockpit panel lights on or off.

(e) LIQ C00L (Liquid Cooling) - The Liquid Cooling Contro1 shall be a green pushbutton switch which shall be used to turn on the Liquid Cooling System when the MCS is not on. The button shall also be lit when the RSC FUNCTION switch is turned to Standby (STBY) or Operate (OPR), and the cooler is energized.

(f) AIR COOL (Air Cooling) - The Air Cooling Control shall be a green pushbutton switch which shall be used to turn on the Blower Assembly when the MCS is not on. The button is also lit when the RSC FUNCTION switch is turned to STBY or OPR, and the blower is energized.

(g) HYD ON (Hydraulics On) - The Hydraulics On Control shall be a green, momentary pushbutton switch which shall be used to turn on the Hydraulic Supply. The Hydraulics On control shall be interlocked with the RSC FUNCTION switch and must be operated each time the RSC FUNCTION switch is turned from OFF to any other position. The Hydraulics On control shall also be interlocked with the Hydraulic Panel of the Antenna Maintenance Fixture, when the Maintenance Fixture is utilized with the Table -Rack. The control shall be operable only when the Hydraulic Panel is in the BENCH position.

(h) HYD OFF (Hydraulic Off) - The Hydraulics Off contro1 shall be a red, momentary pushbutton switch which shall be used to turn off the hydraulics while the MCS is on.

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(i) PHASE SELECTOR - The Phase Selector control shall be a 3-position rotary switch which shall select the phase of the 3-phase, 400-Hz to be measured on the 400-Hz meter. Positions shall be marked A, B, and C.

3.4.1.6.3.3.2.4 Indication. The following monitors and indicating lights shall be provided on the front panel,

(a) RADAR PHASE INDICATOR (400-Hz Phase Selector) - The Radar Phase Indicator shall be a neon indicator light with a translucent white lens which shall show when the 3 -phase, 400-Hz line voltage is connected in the proper phase sequence.

(b) 400 CPS VOLTAGE (AC Voltage Monitor) - The AC Voltage Monitor shall be a voltmeter of 3 percent full scale accuracy which shall measure the MCS primary 400-Hz line voltage between 105 and 125V.

(c) DC VOLTAGE (DC Voltage Monitor) - The DC Voltage Monitor shall be a voltmeter of 3 percent full scale accuracy which shall measure the primary dc line voltage between 22 and 31 Vdc.

(d) ELAPSED TIME (Elapsed Time Indicator) - The Elapsed Time Indicator shall be a meter which conforms to MS 17322-6A. The meter shall indicate total “on time “ of the bench mounted radar.

(e) LIQ FLOW (Liquid Flow Indicator) - The Liquid Flow Indicator shall be a red indicator light which shall show the state of the coolant flow interlock. When the flow through the Liquid Cooling System is less than 2.5+0.1, -0.0 gallons per minute (gpm), the indicator light shall be lit.

(f) HIGH TEMP (High Temperature Indicator) - The High Temperature Indicator shall be a red indicator light which shall show the state of the Temperature High interlock switch. When the return temperature to the Liquid Cooling System rises above 107° ± 3°C (225° ± 5°F) the indicator light shall be lit.

(g) CMPTR AIR (Computer Air Indicator) - The Computer Air Indicator shall be a red indicator light which shall show the state of the Sensing Element installed in the plenum of the Computer Support Structure

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and the state of the Sensing Element installed in the plenum of the Blower Assembly. When the pressure is below nominal 5-1/2 inches of water, in either plenum, the indicator light shall be lit.

(h) MAG TRAP (Magnetic Trap Indicator) - The Magnetic Trap indicator shall be a red indicator light which shall show the state of the Magnetic Trap filter of the Liquid Cooling System. When the trap accumulates sufficient magnetic particles, the indicator light shall be lit to indicate that trap cleaning is necessary.

(i) LOW TEMP (Low Temperature Indicator) - The Low Temperature Indicator shall be a red indicator light which shall show the state of the Temperature Low interlock switch of the Liquid Cooling System. The indicator light shall be lit until the system coolant temperature exceeds 43° ± 3°C (110° ± 5°F) in a rising direction. When this temperature is exceeded, the indicator light shall be off to indicate that the coolant temperature is near the regulated temperature range.

3.4.1.6.3.3.2.5 Protective Devices. The following protective devices shall be provided:

(a) Circuit Breakers - There shall be four pushbutton circuit breakers:

(1) RDR 400 CPS A 3-phase, 50-A breaker for the 115/200V, 400-Hz power to the MCS.

(2) AUX 400 CPS - A 3-phase, 35-A breaker for the 115/200V, 400-Hz power to the table rack.

(3) +28 Vdc - A 70-A breaker for 28 Vdc power.

(4) +28 Vdc STBY (Standby) - A 15 A pushbutton breaker shall be provided for the +28 Vdc standby supply.

(b) Fuse - There shall be a 1-A fuse for the protection of the MCS panel light circuitry.

3.4.1.6.3.3.3 Aircraft Simulator Panel.

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3.4.1.6.3.3.3.1 Function. This panel shall simulate those signals that are normally supplied to the MCS by the aircraft and provide loads where needed or readout lights on those signals from the MCS that normally go to the air - craft.

3.4.1.6.3.3.3.2 Form Factor. This unit shall be 19 inches by 16-1/4 inches by 8-3/4 inches. The weight of the unit shall not exceed 12 pounds.

3.4.1.6.3.3.3.3 Controls. The following controls shall be provided on the panel face. System response to the operation of the controls listed in (c) through (h), (n) and (r) below shall be indicated by illumination of the control itself.

(a) STATION SELECT - The Station Select Control shall be a two-position toggle switch which shall be used to test the missile station select wiring. The positions shall be marked BIT and ACFT (Aircraft).

b) MSL SEL (Missile Select) - The Missile Select Control shall be a three-position toggle switch which shall be used to energize the respective missile circuits in the radar. The positions shall be marked RADAR GUNS and HEAT .

(c) CW STBY (Continuous Wave (CW) Standby) - The CW Standby Control shall be a yellow pushbutton switch which shall be used to place the CW Illuminator of the MCS in the standby mode.

(d) CW ON - The CW On Control shall be a green pushbutton switch which shall be used to turn the CW Illuminator on.

(e) CW OFF - The CW Off Control shall be a red pushbutton switch which shall be used to turn the C W Illuminator off.

(f) XMTR ON (Transmitter ON) - The Transmitter On Control shall be a green pushbutton switch which shall be used to turn the MCS trans-mitter on when the RSC FUNCTION switch is in the OPR position.

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(g) ACCEL (Acceleration) - The Acceleration Control shall be a three -position toggle switch which shall be used to select simulated acceleration signals representative of vertical acceleration of +3g, level flight, or -3g. Positions shall be marked +3, O, and -3.

(h) ALT (Altitude) - The Altitude Control shall be a two-position toggle switch which shall be used to select simulated altitude (see section 3.4.1.6.3.3.3.6 (i) signals representative of the aircraft altitudes to the computer. Positions shall be marked HIGH and LOW,

(i) MBC/GSO INPUTS (Main Beam Clutter/Ground Speed Oscillator Inputs) - A two-position toggle switch shall be provided to switch from normal operation to variable simulation of Main Beam Clutter and Ground Speed.

(j) AIM 7/E (AN/AIM-7E) - The AN/AIM-7E Control shall be a white pushbutton switch which shall be used to activate circuits which generate a simulated missile identity signal (+28Vdc) to the Computer.

(k) MAN MB C (Manual Main B earn Clutter) - A 10-turn potentiometer shall be provided for controlling variable simulation of Main Beam Clutter.

(1) MAN GRD SPEED (Manual Ground Speed) - A ten-turn potentiometer shall be provided “for controlling variable simulation of ground speed,

(m) LAT OFFSET CURSOR (Lateral Offset Cursor Signal) - The Lateral Offset Cursor Signal Control shall be a three-position toggle switch which shall be used to select simulated Lateral Offset Cursor signals of 6.25 miles right (lying on the 30-degree right line on the indicators), O mile (line in the center of the indicators), or 6.25 miles left (lying on the 30-degree left line). Positions shall be marked 30° RIGHT, ZERO, and 30° LEFT;

(n) DATA LINK - The Data Link switch shall be a white pushbutton switch which applies a +28V signal to the Indicator Control Unit of the radar to energize the circuits which enable the display of Data Link signals.

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(o) PLM ACQ (Pilot Lockon Mode-Acquisition) - A five-position, three-pole rotary switch shall provide simulation of the half-action, full-action switch on the pilot's control stick. The switch positions shall be fixed and shall not simulate the spring return feature of the pilot's switch. Two modes of switching shall be provided, a normal mode which shall simulate electrically the pilot's switch and an antenna test mode during which the bore sight signal to the Scan Pattern Generator shall be interrupted in the half-action position so that the antenna remains in the elevated position for checking.

(p) DF (Dogfight) - A single-pole single-throw toggle switch shall simulate the pilot's switch and apply a +28 Vdc control signal to the Tracker, Target Data.

(q) MSL TRIG (Missile Trigger) - A single-pole single-throw toggle switch shall simulate the pilot's missile firing button and apply a +28 Vdc firing signal to the computer firing relay contacts.

{r) SEAM (Sidewinder Extended Acquisition Mode) - A three-position toggle switch shall simulate inputs to the AWG-10 indicators from the AN/ ASA-63 programmer. These consist of a +28Vdc SEAM lock signal to activate the SEAM displays and the azimuth and elevation signals for the aim dot. Two aim dot positions shall be simulated, one at OV, -OV and the other in the lower right quadrant: +6v, -6v.

(s) ANN LAMP (Annunicator Lamp) - A single-pole single-throw toggle switch shall energize all lamps in the Command Indicator via a set of steering diodes as a lamp check.

(t) GEAR INTLK (Gear Interlock) - A single-pole double-throw toggle switch shall simulate switches activated by weight on the aircraft landing gear and the nose wheel retraction and associated relays. The switch shall be appropriately connected to the W/G switch logic circuits in the radar.

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3.4.1.6.3.3.3.4 Indication. The following indicating lights shall be provided on the front panel:

(a) FIRE INTLK (Fire Interlock) - The white Fire Interlock indicator light shall show the presence of the Fire Interlock signal from the Launch Signal Computer.

(b) SPEED GATE - The white Speed Gate indicator light shall show the presence of the Speed Gate signal from the Velocity Signal Computer.

(c) RDR WIND (Radar Wind) - The white Radar Wind indicator light shall show the presence of the Radar Wind Override signal from the Velocity Signal Computer.

(d) ALT # 1 (Altitude No. 1) - The white Altitude No. 1 indicator light shall show the presence of the Altitude Command No. 1 signal from the Launch Signal Compute r.

(e) ALT # 2 (Altitude No. 2) - The white Altitude No. 2 indicator light shall show the presence of the Altitude Command No. 2 signal from the Launch Signal Compute r.

(f) BOOST SP III (AN/AIM-7E Sparrow III Sensing and Band Switch Down) - This white indicator light shall show the presence of the AN/AIM-7E Sensing and Band Switch Down signal from the Launch Signal Computer.

(g) CW RDY (CW Ready) - The green CW Ready indicator light shall indicate that the CW Illuminator's warmup delays have timed out.

(h) CW DLY (CW Delay) - The white CW Delay indicator light shall show the presence of the delayed signal from the CW Illuminator.

(i) XMTR RDY (Transmitter Ready) - The green Transmitter Ready indicator light shall indicate that the transmitter's warmup delays have timed out.

(j) RAD ANT (Radiating from Antenna) - The white Radiation indicator light shall indicate when the system is programmed to radiate out to the antenna.

(k) STOP SCAN - The white Stop Scan indicator light shall show that the antenna is not scanning.

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(1) BIT 7E AND 9D (AN/AIM-7E/9D BIT Command) - The AN/AIM-7E/9D BIT Command indicator light shall show the presence of the AN/AIM-7E/9D BIT command from the Selector-Test Programmer.

3.4.1.6.3.3.3.5 Test Points. Test points shall be provided on the front panel for the following signals from the AN/APG-59:

	Test Point	Test Point Source Impedance
(a)	ENG BIAS (English Bias)	100 kilohms (kn) in parallel with
	AZ (Azimuth)	240 microfarad (mf)
	EL (Elevation)	100kn in paralled with 240 mf
(b)	HEAD AIM	
	H (Roll)	180kn in parallel with 6 mf
	H (Pitch)	180kn in parallel with 6 mf
(c)	SWEEP CONT (Simulated Doppler DC Sweep Control)	Not applicable
(d)	SIM DOPP (Simulated Doppler AC)	Not applicable
(e)	MAN MB C	Not applicable
(f)	VERT DIS POINTER (Vertical Display Pointer)	Not applicable
	OFF BIAS	
	GRD (Ground)	
	PLUS	Connected by 330 n load
	MINUS	
(g)	MAN GRD SPEED	Not applicable
(h)	DIM LAMP SUP (Dimmer Lamp supply)	10 ohms, 5 watts
(i)	CMD IND LAMP TEST VOLTS (Command Indicator Lamp Test Volts)	10 ohms, 5 watts
(j)	RDR IND LAMP TEST VOLTS (Radar Indicator Lamp Test volts)	10 ohms, 5 watts

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(k) WARNING FLAG	1 kn, l/8 watt
(l) SIGNAL GRD (Signal Ground)	Not applicable
(m) SCAN END (Scan Turn Around)	Not applicable
(n) ANT POS (Antenna Position)	Not applicable
AZ (Azimuth)	
EL (Elevation)	
RET (Return)	

3.4.1.6.3.3.3.6 Simulation. The following signals normally provided by the aircraft systerms shall be provided.

(a) Navigational Computer Simulation - A 10 ± lV, 400-Hz, phase-A signal shall be supplied to simulate the North (Vwn) and East (Vwe) Wind Velocity excitation. Load resistors of 360 ohms ± 5 percent, 1/2 watt, shall be provided to simulate the wind signal loads.

(b) Vertical Reference - A three-wire synchro signal for both the pitch and roll inputs to the radar at an angle of O degrees shall be simulated.

(c) Compass Heading - A three-wire synchro signal input to the radar which simulates a O degree heading shall be provided.

(d) Drift Angle - A three-wire synchro signal input to the radar, which simulates a O-degree drift angle shall be provided.

(e) Pressure Ratio (P/PSL) - The panel shall accept an ac input from the AN/APG-59 and provide a simulated pressure ratio by attenuating the input signal with a transfer ratio of O. 673 ± 0.007 to 1.

(f) True Air Speed (TAS) AC - The panel shall accept the TAS (at) . excitation voltage from the Velocity Signal Compute r and simulate a 120O-knot velocity by attenuating the excitation voltage with a transfer ratio of O. 80 ± O. 01 to 1.

(g) TAS (DC) - The panel shall accept the TAS (de) excitation voltage from the Launch Signal Compute r and simulate a 1200-knot velocity by attenuating the excitation voltage with a transfer ratio of O. 80 ± 0.01 to 1.

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(h) Angle of Attack (T) - The panel shall accept a 25V, 400 cycle ac input from the Launch Signal Computer and simulate a 4-degree angle of attack by attenuating the ac input with a transfer ratio of O. 080 ± 0. 002 to 1.

(i) Lateral Offset Cursor - Simulated Later al Offset Cursor signals of 6.25 miles right, 6.25 miles left, and O miles shall be provided for a radar scale factor of 1 mile per volt (de) for selection by the LAT OFFSET CURSOR switch. The source impedance shall be 5000 ohms ± 10 percent,

(j) Main Beam Clutter - A variable Main Beam Clutter (MB C) signal having a magnitude of O to -16 Vdc shall be provided by means of a ten-turn potentiometer Man MBC. This signal shall be fed through a low pass filter having a corner frequency of 210 Hz and shall be selected by MBC/GSO INPUTS switch to replace signals which originate in the Velocity Signal Computer.

(k) Ground Speed - A variable Ground Speed signal having a magnitude of O to -16 Vdc shall be provided by means of a ten-turn potentiometer MAN GRD SPEED, This signal shall be fed through a low-pass filter having a corner frequency of 210 Hz and shall be selected by the MBC/GSO INPUTS switch to replace the signals which originate in the Velocity Signal Computer.

(1) Acceleration - Simulated vertical acceleration signals having a scale factor as specified in MIL-R-81114B shall be provided representing +3 g's, -3 g's, and level flight. Selection of these signals shall be by the ACCEL switch.

(m) Altitude - Two altitude signals of 4.08 ± 0, 0816 Vdc, and 5.26 ± O. 1052 Vdc with scale factors as specified in MIL-R-81114B, shall be simulated. Selection of these signals shall be by the ALT switch.

(n) Climb Angle Simulation - The Panel shall accept a 25V, 400 Hz ac input and shall provide a simulate climb angle signal through a voltage divider ratio of O. 400 ± 0. 004 to 1.

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(o) Data Link - The following test voltages shall be provided for checking the display of signals which normally come from the One Way Data Link.

(1) H	+3.50 Vdc ± 3 percent
(2) Heading Error	-8.73 Vdc ± 3 percent
(3) Target Relative Bearing	-3.50 Vdc ± 3 percent
(4) Range to target	+8.45 Vdc ± 3 percent when loaded with 100k ?

3.4.1.6.3.3.4 MCS Control Panel.

3.4.1.6.3.3.4.1 Function. This panel shall provide the structure and mounting facilities for holding the RSC, the Selector-Test Programmer and the Antenna Hand Contro1 (of 1527 Configuration) of the MCS.

3.4.1.6.3.3.4.2 Form Factor. This unit shall not exceed 19 inches by 11-7/8 inches by 14 inches. The weight of the unit shall not exceed 30 pounds.

3.4.1.6.3.3.4.3 Controls. The controls provided shall be those controls contained on the AN/APG-59 system units installed in the panel, The function of the FUNCTION control of the RSC will be modified as indicated.

(a) When this switch is moved from OFF to STBY, the Liquid Cooling System and Blower Assembly of we Bench shall be energized immediately.. When interlocks on these units are satisfied, the system will be on Standby.

(b) WES - A lighted push-button switch shall be provided to control the power to the Waveguide Extension (WE) 3.4. 3). The light shall be illuminated only if the W ES actually comes on.

3.4.1.6.3.3.5 Manual Built-In-Test (BIT) Panel.

3.4.1.6.3.3.5.1 Function. This panel shall provide the following two functions:

(a) Visual indication of the Selector -Test Programmer output control signals.

(b) Means to initiate commands manually when the Selector-Test Programmer is not being used, and to read out test points for monitoring purposes.

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3.4.1.6.3.3.5.2 Form Factor. This unit shall not exceed 19 inches by 14 inches by 8-3/4 inches. The weight of the unit shall not exceed 20 pounds.

3.4.1.6.3.3.5.3 Controls and Interface. There shall be the following on the panel:

(a) BIT SELECT - The BIT Select control shall be a three-position rotary switch which shall be used to select either the NORM, MAN, or LAMP TEST Mode. When in the MAN mode the Selector-Test Programmer shall be disabled and this panel shall be activated. The LAMP TEST position shall test the lamps. When in the Normal Mode the BIT circuits will be activated by BIT or FIM in response to tape commands. In response to tape command a relay in the Manual BIT Panel shall permit the Test Set, Missile Control System TS-251IC/AWG- 10 to use pushbutton commands of the Manual BIT Panel which will be setin in advance.

(b) Al through E14 - Seventy illuminated white pushbuttons shall be provided to initiate commands when the BIT SELECT control is set to the MAN mode. The lights shall be on to show the presence of a command signal when the BIT SELECT control is set to either the NORM or MAN mode.

(c) MAN BIT TGT (Manual BIT Targets) - A ten-turn potentiometer shall provide variable BIT target signals when the BIT TGT switch is set to the MAN Position. An output from 0 to - 16 Vdc shall be fed through a low-pass filter with a -3 decibel (dB) corner frequency of 21 Hz. Manual targets shall be selected by BIT TGT switch to replace the Digital Analog outputs of the Selector-Test Programmer. The output signal shall be monitored by means of test point MAN BIT TGT.

(d) BIT TGT (BIT Targets Select) - The two-position BIT Targets Select toggle switch shall be used to select either Manual BIT targets or the Normal outputs of the “Selector-Test Programmer. Positions shall be marked MAN and NORM.

(e). TEST POINTS - The following test points shall be provided on the front panel for monitoring purposes.

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(1) BIT READOUT RADAR 1 - It shall permit test probe access to signals from AN/APG-59 Radar Set receiver and transmitter subassemblies that are selected by and routed to the Selector-Test programmer.

(2) BIT READOUT RADAR 2- It shall also permit test probe access to signals from AN/APG-59 Radar Set receiver and transmitter subassemblies, but different from those at BIT Readout Radar 1, which are selected by and routed to the Selector-Test programmer.

(3) BIT READOUT CMPTR 1 (BIT Readout Computer 1) - It shall permit test probe access to signals from the computers that are selected by and routed to the Selector-Test Programmer.

(4) BIT READOUT CMPTR 2 (BIT Readout Computer 2) - It shall also permit test probe access to signals from the computers, but different from these at BIT Readout Computer 1, that are selected by and routed to the Selector-Test Programmer,

(5) GRD (Signal Ground) - It shall provide the signal ground reference for the MAN BIT TGT measurement.

(6) MAN BIT TGT (Variable BIT Targets) - It shall permit test probe access to measure the voltage established by the potentiometer.

(7) COMPARATOR GRD (Selector-Test Programmer Ground) - It shall provide signal ground reference to the Selector Test Programmer when making BIT READOUT measurements.

3.4.1.6.3.3.5.4 Self-Test. Self-test shall be incorporated to check the Al through E14 lights and the power supplies to the Selector-Test Programmer.

(a) Self-test of the lights Al through E14 shall be accomplished by energizing all the lights by means of the BIT SELECT switch.

(b) Self-test of the following power supplies shall be accomplished by means of a green indicator light on the front panel for each of the power supply voltages. When voltage is above 85 percent -O percent + 10 percent of nominal, the associated light shall light.

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(1) -6 Vdc

(2) +6 Vdc Test

(3) +12 Vdc

(4) +25 Vdc

(5) +28 Vdc Suppressed

(6) 28 Vat, ACFT

3.4.1.6.3.3.6 Umbilical Plug and Tuning Drive Panel.

3.4.1.6.3.3.6.1 Function. This panel, in conjunction with the Test Set Missile Control System, TS-2511C/AWG- 10 shall provide the means to test the Umbilical Test Plugs of the Test Set, Umblical Plug TS-2515D/AWM-22. The panel shall also test the four-channel (TG-75/APA-128 Drive or TG-185/AWG-10) and single-channel (TG-76/APA-128 Drive or TG-184/AWG-10) Tuning Drives.

3.4.1.6.3.3.6.2 Form Factor. This unit shall be 19 inches by 15-1/2 inches by 8-3/4 inches. The weight of the unit shall not exceed 19 pounds.

3.4.1.6.3.3.6.3 Controls, The following controls shall be provided on this panel:

(a) CHANNEL SELECT - The Channel Select switch shall be a four-position rotary switch to select any one of four channels of the TG-185/AWG-10 or TG-75/APA-128 Drive, Tuning. The positions of the switch shall be marked 1, 2, 3 and 4.

(b) CIRCUIT BREAKERS - There shall be three pushbutton circuit breakers.

(1) 400 Hz - 115/200 V, 400 Hz, 3-phase, 10 A

(2) 28 Vdc - 10A

(3) LOAD TEST -28 Vdc, 5A

(c) TUN DR SEL - (Tuning Drive Select) - This switch shall be a two-position toggle switch to select between signals from the TG-75 and TG-76 or TG-185 and TG-184 Tuning Drive motors and Firing Interlocks without one loading the other. The two positions of the switch shall be marked SGL CHAN (single channel) and FOUR CHAN (four channel).

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(d) 28 Vdc RESET - This normally closed switch is provided to reset certain relays which lockup when running tests by interrupting the +28 Vdc line.

(e) MNL TEST PRGM (Manual Test Program) - This switch shall be a 30-position thumb -wheel switch with two thumb wheels and three modules. The switch shall be capable of setting up, in proper sequence, the various test conditions to check for proper operation of any of the four Tuning Drives.

(f) AIM-7F - This single-pole double -throw toggle switch shall apply at 10. 2V signal to the limit output in the ON position, and other voltages (3.4.1.6.3.3.6.6.1) shall be applied in the OFF position as selected by other switching.

(g) PD ILL (pulse Doppler Illuminate) - This single-pole single-throw toggle switch shall apply +28 Vdc to the +28 Vdc delayed output line in the ON position, and in the OFF position shall be an open circuit.

(h) NO GO - RECYCLE - This single-pole double -throw toggle switch shall select the 3. 2 Vdc signal at the recycle line in the ON position, and other voltages (3.4.1.6.3.3.6.6.3) shall be applied in the OFF position as selected by other switching.

(i) NO GO - AFC/LIM (NO GO-AFC/Limit) - This double -pole double-throw toggle switch, in conjunction with other switching, shall apply the 1.4V NO GO signals to the AFC line and the 7.9V NO GO signal to the limit line in the ON position. In the OFF position, other switching shall select the signal, on this line (3.4. 1. 6.3.3.6.6.1 and 3.4.1.6.3. 3.6.6.2).

3.4.1.6.3.3.6.4 Indicator Lights. The following indicator lights shall be provided on the front panel.

(a) TUNE-UP - A white TUNE-UP indicator light shall show that the selected Klystron Tuning Motor Signal is +28 Vdc for a TUNE-UP command

(b) TUNE-DOWN - A white TUNE-DOWN indicator light shall show that the selected Klystron Tuning Motor Signal is ground for a TUNE-DOWN command.

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(c) FIRE INTLK (Fire Interlock) - A white Fire Interlock indicator light shall show that the tuner has been tuned and has switched to Automatic Frequency Control (AFC).

(d) GO - A green GO indicator shall indicate when continuity exists during the check of the five ground tests on the Tuning Drives. It shall also light during the checks of the roll commands when the inputs are within the proper voltage ranges.

3.4.1.6.3.3.6.5 Electrical Connectors. Connections to the units under test shall be as follows:

Reference Designation	Function
J1	Input Power
J2	Motor Fire, Wing
J3	Motor Fire, Fuselage
J4	AN/AIM-7D/7E, Umbilical
J5	AN/AIM-9B/9D, Umbilical
J6	Test Set Missile Control System, TS 251lC/AWG-10
J7	Drive, Tuning TG-75/APA-128 or TG-185/AWG-10
J8	Drive, Tuning TG-75/APA-128 or TG-185/AWG-10
J9	Drive, Tuning TG-76/APA-128 or TG-184/AWG-10
J10	Cartridge Firing Input
Jll	AN/AIM-9B/9D, Motor Fire
J12	Test Point: HOLD
J13	Test Point: LIM (Limit)
J14	‘Test Point: AFC
J15	Test Point: RECYCLE
J16	Test Point: FIRE INTLK (Fire Interlock)

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Reference Designation	Function
J17	Test Point: TUN MOT DR (Tuning Motor Drive)
J18	Test Point: STBD-ROLL CMD (Starboard Roll Command
J19	Test Point: PORT ROLL CMD (Port Roll Command)
J20	Test Point: 28V DELAYED
J21	Test Point: SIG GND (Signal Ground)

3.4.1.6.3.3.6.6 Test Outputs. The Test Panel shall provide the output simulations listed below.

3.4.1.6.3.3.6.6.1 Limit Voltage. The following voltages shall be simulated from a 12kn ± 10 percent source impedance. The MNL TEST PRGM switch, and other switches, shall select the voltage to be supplied.

(a) +4. 1 *0. 08 Vdc for AIM-7D/7E series of missiles

(b) -4.1 ± 0. 08 Vdc for AIM-7D/7E series of missiles

(c) +7.9 ± 0. 16 Vdc for upper limit NO GO test

(d) -7.9 ± 0. 16 Vdc for upper limit NO GO test

(e) +10.2 ± 0. 2 Vdc for AIM-7F series of missiles

3.4.1.6.3.3.6.6.2 AFC Voltages. The following voltages shall be simulated from a 75kn ± 10 percent source" impedance. The MNL TEST PRGM and AFC/LIMIT NO GO switches shall select the one to be applied.

(a) +1.4 + 0.06, -0 Vdc (increasing from 0)

(b) -(1. 4 + 0.06, - 0) Vdc (increasing from 0)

(c) +2.2 ± 0. 04 Vdc (increasing from 0)

(d) -2.2 ± 0. 04 Vdc (increasing from 0)

(e) +1.0 + 0.04, -0 Vdc (decreasing toward 0)

(f) -(1.0+0.04, - 0) Vdc (decreasing toward 0)

3.4.1.6.3.3.6.6.3 Recycle. The voltages shall be simulated from a source impedance of 150kn ± 10 percent. The MNL TEXT PRGM and Recycle NO GO switches shall select which is to be applied.

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(a) +3. 2 ± 0.06 Vdc (increasing from 0)

(b) +3. 8 + 0, -0.19 Vdc (increasing from 0)

(c) +2.7 + 0.05, -0 Vdc (decreasing toward 0)

3.4.1.6.3.3.6.6.4 28 Vdc Delayed. A +28 Vdc signal shall be present at this output when the PD ILL switch is ON.

3.4.1.6.3.3.6.6.5 28 Vdc Hold. When the MNL TEST PRGM switch reaches the appropriate step in the test sequence, +28 Vdc shall be applied at this output. A latching circuit shall be provided which will keep the signal on for subsequent steps until the 28 Vdc RESET switch is momentarily depressed.

3.4. 1.6.3.3.6.7 Test Inputs. The Test Panel shall provide for monitoring the following inputs coming back from the Tuning Drives in response to the switching and outputs described in 3.4.1.6.3.3.6.3 thru 3.4.1.6.3.3.6.6 3.4.1.6.3.3.6.7, 1 Roll Channels. The Test Panel shall verify that the Right Roll command is +18.8 Vdc ± 10 percent and that the Left Roll command is -18. 8 Vdc ± 10 percent when checking four-channel tuners.

3.4.1.6.3.3.6.7.2 Tuning. Provisions shall be made for indicating which of the following five conditions exists on each of the tuning inputs.

(a) TUNE-UP: +28 Vdc

(b) TUNE-UP with Recycle: +28 Vdc through 180 ohms

(c) TUNE-DOWN: ground

(d) TUNE-DOWN with Recycle: ground through 180 ohms

(e) Open Circuit

3.4. 1.6.3.3.6.7.3 Grounds. Provisions shall be made for checking all the separate ground circuits for the single - and four -channel tuners.

3.4.1.6.3.3.6.7.4 Firing Interlocks. Presence/absence of the +28 Vdc Firing Interlock voltage shall be monitored for the selected missile channel.

3.4.1.6.3.3.6.8 Cable. The J7, J8, and J9 connectors shall be part of cables wired directly to the test panel. This cable shall enable any of the four

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Tuning Drives to be connected to the panel while resting on the drop leaf of the Receiver Section Bench Assembly. The cable shall retract into the panel when not in use.

3.4.1.6.3.3.7 Voltage Regulator.

3.4.1.6.3.3.7.1 Function. An ac Voltage Regulator shall be provided for regulating the 115-V, 400-Hz, Phase -A which is used by Tuning Drives TG-75/AFA-128 and TG-76/APA-128 or TG-185/AWG-10 and TG-184/AWG-10. This regulator shall maintain the 400 -Hz power at 115 V ± 1 percent when the line voltage varies between 107 and 120 V. The current carrying capability of the regulator shall be 1A minimum.

3.4.1.6.3.3.7.2 Form Factor. The Voltage Regulator shall be hermetically sealed and shall not exceed 6-1/2 inches by 3- 1/2 inches by 5 inches. It shall be bracket mounted on the inside right of the Test and Control Rack behind the MCS Control Panel. The weight of the unit shall be approximately 6 pounds.

3.4.1.6.3.3.8 IFF Simulator Panel.*2

3.4.1.6.3.3.8.1 Function. The IFF Simulator Panel shall provide test points, circuits, and indicator lights to verify proper operation of the SEA circuits.

3.4.1.6.3.3.8.2 Form Factor. This unit shall be 10 x 4 x 6 inches and designed to mount against the upper right side of the Test and Control Rack. It shall not weigh more than 5 pounds.

3.4.1.6.3.3.8.3 Controls. The following controls shall be on the unit:

(a) IFF SIMULATION - This switch shall be a DPDT toggle switch for selecting whether the SEA tests shall be controlled from the Table Rack or the Selector Test Programmer, and together with associated relays permit exercising the operation enable, video #1, and IFF display video lines.

(b) INTEROG TEST (Interrogate Test) - This switch shall be a momentary pushbutton switch to apply +28 Vdc to the Interrogate # 1 circuit when this is required.

*2 see paragraph 6.5.2

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3.4.1.6.3.3.8.4 Indicator Lights. The following white Indicator Lights shall be mounted on the front panel to indicate when the respective circuits in the radar are energized with +28 Vdc.

(a) INTEGROG #1

(b) INTEGROG #2

(c) T C CMD

(d) LONG RANGE DISPLAY ENABLE

3.4.1.6.3.3.8.5 Test Points. The following front panel test points shall provide a monitoring point for the respective signals within the radar:

(a) PRE TRIGGER #l - The cable for this signal shall be terminated in 91 ohms and 100 pF.

(b) RDR TRIGGER (Radar Trigger) - The cable for this signal shall be terminated in 91 ohms.

(c) PULSE RDR VIDEO (Pulse Radar Video) - The cable for this signal shall be terminated in 91 ohms and 100 pF

(d) t_0 PULSE - The cable for this signal shall be terminated in 91 ohms and 100 pF.

(e) WARNING PULSE - The cable for this signal shall be terminated in 91 ohms and 100 pF.

(f) EARLY R GATE (Early Range Gate) - The cable for this signal shall be either terminated in 91 ohms and 100 pF or returned on the video lines as they are exercised per 3.4.1.6.3.3.8.3 (a).

(g) SIGNAL GRD

3.4.1.6.3.3.8.6 Test Signal. A test signal shall be provided as follows:

(a) EL STROBE JITTER - This signal shall be provided from a -6 Vdc input through a 15 kohm ± 1%, 1/8 W resistor.

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3.4.1.6.4 Indicator Bench.

3.4.1.6.4.1 Function, The Indicator Bench shall mount and furnish an operating environment for the AN/APG-59 Radar Set Command indicator, Radar Indicator, and Indicator Control, and provide 115 -Vat, Convenience Power Outlets, conforming to MIL-C-3767.

3.4.1.6.4.2 Form Factor. The Indicator Bench shall not exceed 64 inches by 28- 1/8 inches by 75 inches and the weight shall not exceed 550 pounds. The Indicator Bench shall consist of the following items:

(a) Mechanical Support Structure.

(b) Radar Indicator Support Structure.

(c) Command Indicator Support Structure.

(d) Indicator Control Support Structure.

(e) Test Set Support Structure.

3.4.1.6.4.2.1 Mechanical Support Structure.

3.4.1.6.4.2.1. 1 Function. The Mechanical Support Structure shall be capable of being bolted to the deck. It shall mount the items specified in (see section 3.4.1.6.4.2(b), (c) and (d)) and shall provide work and storage space.

3.4.1.6.4.2.1.2 Form Factor. The Mechanical Support Structure shall provide a bench top a nominal 64 inches by 26 inches, 31 -7/8 inches above deck level. Storage space shall be provided beneath the bench top in the form of drawers and cabinet space. A vertical storage drawer shall be installed at the center of the Mechanical Support Structure below the bench top to store cables used with the Table-Rack. The Radar Indicator Support Structure shall be installed on the right side of the bench top, The Command Indicator Support Structure shall be installed on the left side of the bench top . The top rear of the Mechanical Support Structure shall form a shelf 64 inches by 15 inches whose upper surface shall be 75 inches above the surface of the deck, Provisions shall. be made to suspend the Indicator Control Unit of the AN/APG-59 Radar Set below the top shelf at the right side of the Mechanical Support Structure and the Test Set Support Structure below the top shelf at the left side of the Mechanical Support Structure.

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3.4.1.6.4.2.1.3 Convenience Power Outlets. Four Duplex Convenience Power Outlets rated at 115 Vat, 25A shall be provided at the rear of the structure above the Bench Top, These Outlets shall receive their power from the convenience power outlet terminal strip installed on the underside of the center section of the upper shelf, The Outlets shall be 2-pole, 3-wire, shall be polarized and shall meet the requirements of MIL-C-3767, The terminal strip shall be connected directly to the Primary ac Power Source by the using facility and shall not be connected through the Circuit Breakers.

3.4.1.6.4.2.2 Radar Indicator Support Structure.

3.4.1.6.4.2.2. 1 Function. This structure shall support the Radar Indicator s o that it can be viewed easily from several angles and can be rotated and raised for each maintenance.

3.4.1.6.4.2.2 Form Factor. The normal mounting points of the Radar Indicator shall be used, and the Structure shall incorporate a turntable capable of rotating and locking the Radar Indicator at any angular position within ± 90 degrees ± 5 degrees in the horizontal plane. The Structure shall be capable of lifting and locking the Radar Indicator at 30 degrees elevation and also to be slid forward and locked in the vertical position to provide access to the bottom of the indicator. The weight of this structure shall not exceed 38 pounds.

3.4.1. 6.4. 2.3 Command Indicator Support Structure.

3.4.1. 6. 4.2.3.1 Function. The Command Indicator Support Structure shall support the Command Indicator and shall make it accessible for repair.

3.4.1.6.4.2.3.2 Form Factor. The normal mounting points of the Command Indicator shall be utilized. The structure shall be capable of being lifted and locked at 30 degrees in the vertical plane. The structure shall also have slide rails to permit the Command Indicator being slid to the front of the Indicator Test Bench and then to be lifted and locked at 90 degrees elevation and also to be slid forward and locked in the vertical position to provide access to the bottom of the indicator. The weight of the support structure shall not exceed 14 pounds.

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3.4.1.6.4.2.4 Indicator Control Support Structure.

3.4.1.6.4.2.4.1 Function. This structure shall support the Indicator Control and shall be attached to the lower surface of the overhead shelf.

3.4.1.6.4.2.4.2 Form Factor. This unit shall fasten to the underside of the shelf and support the indicator Control from its normal mounting fixtures. Nominal dimensions shall be 17 inches by 8 inches by 12 inches. The weight of this unit shall not exceed 8 pounds.

3.4.1.6.4.2.5 Test Set Support Structure.

3.4.1.6.4.2.5.1 Function. The Test Set Support Structure shall support the MCS Test Group OQ-50A/AWM-21 below the top shelf at the left side of the Mechanical Support Structure when the Test Set is used to perform tests on LRU's of the AN/APG-59 Radar Set which are normally serviced at the Indicator Test Bench and the Computer Test Bench. The structure shall support the Test Set so that it can be viewed from several angles in elevation and azimuth for ease in use.

3.4.1.6.4.2.5.2 Form Factor. The Test Set Support Structure shall not exceed 21 inches by 12 inches by 16 inches and shall weigh not more than 25 pounds. The structure shall be in the form of a yoke assembly designed to permit rotation and locking of the Test Set in elevation and azimuth.

3.4.1.6.5 Computer Bench.

3.4.1.6.5.1 Function. The Computer Bench shall mount and furnish an operating environment for the AN/APG-59 Radar Set Launch, Pilot Command and Velocity Signal Computers, and provide suitable working space for necessary testing and repair.

3.4.1.6.5.2 Form Factor. The Computer Bench shall not exceed 28- 1/8 inches by 28 inches by 75 inches and the weight shall not exceed 225 pounds. The Computer Bench shall consist of the following items:

(a) Mechanical Support Structure.

(b) Computer Support Structure.

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3.4.1.6.5.2.1 Mechanical Support Structure.

3.4.1.6.5.2.1.1 Function. The Mechanical Support Structure shall mount the Computer Support Structure and shall provide work and storage space.

3.4.1.6.5.2.1.2 Form Factor. The Mechanical Support Structure shall be capable of being bolted to the deck and shall provide nominal 28 inches by 28 inches bench top. Storage space shall be provided.beneath the bench top in the form of drawers and cabinet space. The Computer Support Structure shall be installed on the bench top of the Mechanical Support Structure. The top rear of the Mechanical Support Structure shall form a shelf 28 inches whose upper surface shall be 75 inches above the surface of the deck.

3.4.1.6. 5.2.1.3 Control. An EMER OFF switch shall be mounted to the f rent surface of the shelf on an appropriate bracket. The EMER OFF switch shall activate the same circuits as the one specified in 3.4.1.6.2.8.3.

3.4.1.6.5.2.2 Computer Support Structure.

3.4.1.6.5.2.2.1 Function. The Computer Support Structure shall provide mounting for the AN/APG-59 Radar Set Velocity, Launch, and Pilot Command Signal Computers and shall distribute cooling air to them so that they can be operated and maintained.

3.4.1.6.5.2.2.2 Form Factor. The structure shall consist of a four-sided vertical plenum chamber mounted on a swivel base capable of being rotated ± 85 ± 5 degrees in azimuth. One side of the plenum chamber shall contain an air inlet and filter to supply ambient room air to the blower. The three signal computers shall mount to the three sides of the plenum chamber,

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each computer being assigned to a given side of the chamber. The computers shall be attached to the plenum by normal mounting points, and shall receive cooling air distributed through ports in the side of the plenum. Maximum dimensions of the Computer Support Structure shall be 17 inches by 17 inches by 21-1/2 inches, with plenum covers installed. The weight of the Computer Support Structure including plenum covers shall not exceed 60 pounds.

3.4.1.6.5.2.2.3 Performance. The Computer Support Structure shall deliver to the computers a minimum of 1.25 pounds per minute of air at a pressure of 5 inches of water.

3.4.1.6.5.2.2.4 Primary Input Power Requirements. The Computer Sup - port Structure Primary Input Power Requirements shall be 3 -phase, 103.5 to 126. 5-Vac, 380 to 420-Hz, 50-VA per phase.

3.4.1.6.5.2.2.5 Contents. The Computer Support Structure shall contain the following elements with indicated characteristics,

(a) Air Filter - The Air Filter shall be oil-treated and reusable. The Filter shall have a ventilation area of 32 square inches.

(b) Blower - The Blower shall be a vaneaxial fan with an integral, 3-phase, 4-wire, 1 15/200 V, 400-Hz motor. The Blower shall deliver air into the plenum per performance (see section 3.4.1.6.5.2.2. 3).

(c) Sensing Element - A differential pressure switch shall mount to the plenum chamber. It shall be of the diaphragm type and shall sense static pressure differential between the air plenum and outside ambient air. The switch shall activate when the air pressure decreases to 5.5O; -0.00, +0.25 inches of water. The output of the sensing circuit shall be electrically connected to the CMPTR AIR indicator, in series with a similar Sensing Element installed in the plenum chamber of the blower assembly.

(d) Plenum Covers - Three removable plenum covers, one for each computer unit, shall be provided for the three plenum surfaces. These covers shall block the computer cooling air ports in the absence of a computer in order to prevent the loss of cooling air pressure.

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3.4. 1.6.6 Hydraulic Supply.

3.4. 1.6.6.1 Function. The Hydraulic Supply HD-776A/AWG-10 shall provide a single hydraulic pressure capability and shall provide hydraulic power to the Antenna Maintenance Fixture or to the Antenna Support Fixture with the table -rack for operation of the antenna.

3.4.1.6.6.2 Form Factor. The Hydraulic Supply shall be of rectangular configuration, Maximum dimensions exclusive of hoses, fittings and connectors, shall be 24-3/4 inches by 17-7/8 inches by 10-3/4 inches. It shall have a control panel recessed from the top of the unit containing operating controls and indicators. Two nonprotruding handles shall be provided on the top surface and shall be spring loaded to return to stowed position. The weight of this unit including oil shall not exceed 80 pounds. This unit shall be mounted in the base of the Anten`na Support Fixture and the operation normally controlled from the Power Control Panel through the table -rack interconnecting harness. The unit shall also be capable of delivering rated Performance when mounted at a remote station up to 40 feed distant from the AN/APG-59 hydraulic connections. Electrical cabling and hydraulic hoses for remote mounting are not furnished as part of the Radar Maintenance Table-Rack.

3.4.1.6.6.3 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or a reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment.

3.4.1.6.6.4 Primary Input Power Requirements. The equipment power requirement shall be:

AC Power: 3-phase, 4-wire 107.5 to 119.5 V, phase to neutral, 380 to 420 Hz, 2800 VA.

3.4. 1.6.6.5 Noise Level. The noise level shall not exceed 70 dB above reference level of O. 0002 microbar when measured 6 feet from the unit.

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3.4.1. 6.6.6 Performance. The output of the Hydraulic Supply Unit shall be the following:

(a) Pressure - The pressure supply shall be at least 880 pounds per square inch, gage (psig) but not greater than 1100 psig.

(b) Flow Rate - The steady-state flow rate shall be not less than 1.5 gpm at 975 psig. The flow rate shall be reducible by bypassing internally all flow while maintaining an outlet pressure of less than 1025 psig,

(c) Hydraulic Fluid - The unit shall operate with hydraulic fluid per MIL-H-5606.

(d) Temperature - The unit shall provide cooling to maintain the temperature of the fluid supply at 54 ± 17°C(130+300F).

3.4.1.6.6.7 Contents. This unit shall consist of the following with indicated characteristics.

(a) Pump - A constant displacement, motor-driver pump designed for use with MIL-H-5606 hydraulic fluid shall be provided. The pump shall be capable of providing pressures and flow rates consistent with applicable Performance when power is applied.

(b) Filters -

(1) A 10 -micron absolute, metal mesh filter of 6 gpm capacity shall be in series with the pressure supply line. The filter shall be equipped with an automatic reset differential pressure device which will cause the FILTER indicator to light when the pressure drop across the filter element reaches 80 ± 12 psi.

(2) A 105-micron absolute, metal mesh filter of 6 gpm capacity shall be in series with the return line.

(c) Reservoir - A reservoir with a minimum capacity of 1.0 gallons shall be provided. The reservoir shall have two-way vent valve constructed so that spillage will not occur in any attitude. The fill opening shall have a removable 300 -micron strainer and a sealing cap. A Fluid Level Indicator shall monitor the quantity of fluid in the reservoir.

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(d) Heat Exchanger - A heat exchanger shall be installed in the return line. The heat exchanger shall be a liquid-to-air type and shall maintain the fluid temperature at 54.4° ± 16.6°C (130° ± 30°F) through use of an air blower and thermostatically controlled bypass.

3.4.1.6.6.8 Controls. The following controls shall be provided on the Control Panel.

(a) START-ON - The Start-On control shall be a flush square, illuminated panel push switch conforming to MIL-S-22885. When external power is applied to the unit by the HYD-ON switch the legend START shall be illuminated in black letters on white background, The pumping unit shall be energized by depressing the illuminated panel. When the illuminated panel is depressed, the legend ON shall be illuminated in black letters on green background and the Start Legend shall be deenergized.

(b) STOP-OFF - The STOP-OFF control shall be a flush square, illuminated panel push switch conforming to MIL-S-22885. When external power is applied to the unit by the HYD-ON switch the legend OFF shall be illuminated in black leters on a red background. When the START -ON switch is depressed, the OFF legend shall be deenergized and the legend STOP shall be illuminated in black leters on a white background. Depressing the switch will deenergize the pumping unit.

(c) LAMPS (Lamp Control Switch) - The Lamp Control Switch shall be a single pole, single throw toggle switch conforming to MS 25307-252 to energize or reenergize all switch and indicating lamps.

(d) LAMP TEST - The LAMP TEST control shall be a flush square, illuminated panel push switch conforming to MIL-S-22885 to test the filter, pump and pressure indicating lamps and the stop and on switch lights. When external power is applied to the unit by the HYD -ON switch the legend LAMP TEST shall be illuminated in black letters on a white background and shall remain illuminated as long as external power is applied to the unit and the Lamp Control Switch is on.

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3.4.1.6.6.9 Panel Indicators. The following monitors and indicators shall be provided on the control panel.

(a) FAULT-PRESS (Pressure) - The pressure indicating lamp shall light when the pumping unit supply pressure is below normal operating value. The lamp lens shall be red with the legend PRESS in black letters.

(b) FAULT-FILTER - the filter indicating lamp shall light when the pressure drop across the filter element reaches 80 ± 12 psi. The lamp lens shall be red with the legend FILTER in black letters.

(C) FAULT-PUMP - The pump indicating lamp shall light when the inlet pressure to the pump is below the minimum required inlet pressure. The lamp lens shall be red with the legend PUMP in black letter.

(d) SUPPLY PRESSURE - A 0-2000 psi Pressure Gauge shall be provided to monitor the supply line pressure downstream of all other components. Major graduations shall be a 0, 1000 and 2000 psi with minor graduations every 100 psi. The gauge shall have an accuracy of 3 percent of full scale reading.

(e) HOURS - An elapsed Time Meter per MS17322 shall be provided to indicate total operation time of the Hydraulic Supply Pump and Blower.

(f) Fluid Level Indicator - A gauge shall be provided to indicate the Hydraulic fluid level. The reservoir fluid level shall be visible from outside the unit when the reservoir is from 1/3 to 3/4 full.

(g) AC C- PRESS (Accumulator Pressure) - When the nitrogen pressure in the accumulator is less than 380 psig and power is applied to start the unit, a red indicator shall be illuminated.

3.4.1.6.6.10 Protective Devices.

3.4.1.6.6.10.1 CIRCUIT BREAKER. A Circuit Breaker shall be provided suitable for 115V phase to neutral, 200V, phase to phase, 3-phase, 400-Hz operation. It shall be a mechanically coupled, 3-pole, manual reset type and shall have a maximum continuous rating of at least 10 A.

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3.4.1.6.6.10.2 Reservoir Level Switch. A switch shall be provided, interlocking the pump motor circuit, to monitor the fluid level within the Reservoir and the attitude of the Hydraulic Supply. The switch shall open and interrupt one pump motor circuit to protect the pump from running without any fluid at any combination of level and inclination.

3.4.1.6.6.10.3 Pressure Switch. A normally open switch shall be located in the pressure line and shall close on a rising pressure at 950 ± 20 psig. It shall open on decreasing pressure at 865 ± 55 psig. The pressure switch closing shall cause the FAULT-PRESS indicator to be lit.

3.4.1.6.6.10.4 Accumulator Pressure Switch. A pressure switch connected to the accumulator shall prevent the unit from starting, but cause illumination of a fault indicator, if the static nitrogen pressure in the accumulator drops below 450 +0, -50 psig.

3.4.1.6.6.11 Connections.

3.4.1.6.6.11.1 Electrical. A primary power input receptacle per MS3102R28-19P shall be provided on the Control Pane1. Connection shall be made from the Power Contro1 Panel.

3.4. 1.6.6.11.2 Hydraulic. Two Hydraulic connections shall be located on the control panel and shall be quick discount types. The pressure SUPPLY coupling and the RETURN line coupling shall mate with the hoses.

3.4.1.6.6.12 Hoses. Two medium-pressure hydraulic hoses, not exceeding 67 inches in length, shall be provided to connect the Hydraulic Power Supply to the Antenna or the Transfer Valve Assembly of the Antenna Maintenance Fixture. Both hoses shall be equipped with quick-disconnect connectors and shall be constructed with an extruded Teflon resin liner and corrosion resistant wire braid.

3.4.1.6.7 Liquid Cooling System.

3.4. 1.6.7.1 Function. The Liquid Cooling System shall provide OS25 liquid coolant to the AN/APG-59. The system shall pump, cool, and regulate coolant fluid. and shall operate as either a liquid-to-air or liquid-to-sea water heat exchanger.

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3.4.1.6.7.2 Form Factor. The Liquid Cooling System shall be housed in an enclosure 20 inches by 43 inches by 20 inches. Four flush-mounted carrying handles, two on each side, shall be provided. A recessed control panel on the top surface of the unit shall contain all necessary connections and controls for operation. Weight, excluding hoses" and coolant, shall not exceed 400 pounds. The Liquid Cooling System shall be mounted on the deck adjacent to the Antenna Support Fixture or at a remote station located no more than 40 feet distant from the AN/APG-59 Radar Set coolant connections.

3.4.1.6.7.3 Primary Input Power Requirements. - The equipment power requirements shall be 3-phase, 4-wire, 115/200 Vat, 400 Hz, 1600 VA.

3.4.1.6.7.4 Sea Water Requirements. - The liquid-to-sea water heat exchanger shall be capable of the heat transfer specified under performance with supplied sea water having the following characteristics:

(a) Pressure - Supply pressure shall-be 175 psig maximum.

(b) Temperature - Maximum input temperature shall be 320 C (90°F).

(c) Flow Rate . Flow rate shall be 5 ± 0.5 gpm.

3.4. 1.6.7.5 Noise Level. The acoustical noise level produced by this equipment shall not exceed 80 dB (0. 0002 microbar reference) when measured 6 feet from the unit.

3.4.1.6.7.6 Performance. When operating as complete system, the Liquid Cooling Systerm shall be capable. of removing 10,700 watts of heat from the coolant fluid and shall provide the following:

(a) Flow Rate - Liquid coolant flow rate shall be 2.7 gpm minimum.

(b) Pressure - The pumping" system shall be capable of producing the rated flow with an external pressure drop of 135 psi maximum from supply to return connections with a pressure at the return connector which shall not exceed 35 psig. The supply pressure shall not exceed 225 psig under all flow conditions.

(c) Temperature - The output temperature shall be maintained to 52° ± 5. 5° C (125 ± 10° F) except when the inlet fluid temperature shall fall below 49° C (120° F).

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3.4.1.6.7.7 Contents of Unit. The Liquid Cooling System shall consist of the following:

3.4.1.6.7.7.1 Pump Motor Assembly. There shall be a motor-pump assembly capable of circulating the coolant as specified in 3.4.1.6.7.6. There shall be a fan either as part of this Assembly or driven by separate motor to circulate ambient air through the unit when the Liquid-to-Air Heat Exchanger is in use. The air flow shall be sufficient to meet performance specified in 3.4.1.6.7.6 under conditions as specified in section 3.4.1.6.7.7.2.

3.4.1.6.7.7.2 Liquid -to-Air Heat Exchanger. This Exchanger shall be capable of heat transfer as specified in 3.4.1.6.7.6 with ambient air temperature and altitude profile as follows:

(a) 40°C (104°F) from sea level to 8000 feet.

(b) Linear decrease from 40°C (104°F) at 8000 feet to 35°C (94°F) at 10, 000 feet.

3.4.1.6.7.3 Liquid-to-Sea Water Heat Exchanger. This Exchanger shall be capable of heat transfer specified in 3.4.1.6.7.6 with sea water available as specified in 3.4.1.6.7.4. A replaceable zinc anode plug shall be provided on the sea water side.

3.4.1.6.7.7.4 Reservoir. A 2.5 gallon minimum capacity reservoir shall be provided with sight glass and relief valve. The relief valve shall be in the supply pressure line and shall limit the liquid coolant delivered to the AN/APG-59 Radar Set to a maximum pressure of 225 psig. The sight glass shall be graduated with at least 1/4, 1/2, 3/4, and FULL markings.

3.4.1.6.7.7.5 Control Panel. A recessed control panel shall be provided on the top surface of the enclosure. The panel shall mount all necessary external connections, consisting of coolant quick-disconnect couplings, reservoir filler with cap, sea water fittings, and an electrical power receptacle. A selector value and an ELASPED TIME indicator conforming to MS 17321-2 shall also be provided on this panel.

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3.4.1.6.7.7.5.1 Connections. The following connections shall be provided on the Control--Panel:

(a) Coolant .- A SUPPLY and RETURN quick-disconnect connector shall be provided to mate with coolant lines specified in 3.4.1.6.7.7.6.

(b) Sea Water - An OUT and an IN water connector shall each be supplied. Connectors shall be l/2-inch outside diameter tube, stainless steel, fittings with 3/4 -16 UNF-24 standard thread.

(c) Electrical - An electrical recptacle marked 63A1A3J1 or 63A7J1 shall be provided to mate with MS3126-F-14-12S.

3.4.1.6.7.7.6 Coolant Lines. Two 5/8-inch diameter lines 40 feet ± 1 foot long with quick-disconnect fittings at each end shall be provided. Connectors shall be MS33656.

3.4.1.6.7.8 Controls. The selection of air or sea water as the heat exchanger medium shall be made by a manual selector valve located on the Control Panel. Positions shall be marked WATER and AIR. Electrical power shall be applied through the LIQ COOL pushbutton control.

3.4.1.6.7.9 Protective Devices.

3.4.1.6.7.9.1 Safety Interlocks. The following Safety Interlocks shall be provided:

(a) Temperature High - A temperature switch to allow interlocking of the MCS 28 Vdc standby and operate signal with the fluid temperature shall be interrupted when the RETURN Coolant connector exceeds 1070 ± 3°C (225° ± 5°F) on rising temperature. It shall close at 99°C (210°F) minimum on a falling temperature. The contacts shall be rated for 2A at 28 Vdc. The HIGH TEMP indicator shall be illuminated when the switch is open.

(b) Temperature Low - A temperature switch shall allow +28 Vdc to be applied to the LOW TEMP indicator to show when the coolant temperature is below the regulated temperature range. The switch shall be interrupted when the temperature of the coolant at the SUPPLY coolant connector reaches 43° ± 3°C (110° ± 5°F) in a rising direction and shall close at 32°C (+90°F) minimum on a failing temperature, The contacts shall be rated for 2 A at 28 Vdc.

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(c) Flow - A flow switch shall allow interlocking of the MCS 28 Vdc standby and operate signal with the fluid flow. It shall be closed when the fluid flow exceeds 2.75 gpm and shall be interrupted when the flow falls below 2.5 gpm. The contacts shall be rated for 2A at 28 Vdc. The LIQ FLOW indicator shall be illuminated when the switch is open.

3.4.1.6.7.9.2 Filters..

(a) Magnetic Trap - A Magnetic Trap shall be provided in the coolant fluid supply line which shall accumulate magnetic particles from the fluid. When the Trap collects magnetic inpurities sufficient to close an electrical circuit, the Magnetic Trap (MAC TRAP) indicator shall be lighted.

(b) Coolant Fluid - A nominal 30 -micron filter shall be provided in the supply line.

(c) Sea Water - A bronze sea water strainer of 50 to 125 micron size shall be provided in the inlet line.

3.4.1.6.8 Test Accessories.

3.4.1.6.8.1 Cable Installation. The Cable Installation shall be a group of cables, adapters, fittings, and associated hardware to provide the facilities for connecting the Analyzer, Radio Frequency Noise (RFNA) to the MCS in the various ways and combinations as required to perform the analysis of MCS operation by the RFNA. The Cable Assembly shall be in the form of a kit delivered as one of the Test Accessories furnished with the table - rack, Radar Maintenance. Cables shall be provided as necessary to make all the test setups for each measurement to be made by the RFNA (under the condition that any cable may be used in more than one setup whenever feasible). Any and all adapters need to match the cable(s) to the appropriate junctions on the MCS and/or the RFNA shall be provided. All connectors on the cables shall mate with the appropriate connectors. All cables shall be of sufficient length to reach between connection points in a standard sidebench installation but within permissable limits of attenuation.

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3.4.1.6.8.2 Modulator Test Cover Assembly.

3.4.1.6.8.2.1 Function. The Modulator Test Cover Assembly shall replace the Modulator Housing Cove r during testing and maintenance of the RF Section of the Radar Set. The Test Cover shall provide safe access to the units within the Modulator Housing.

3.4.1.6.8.2.2 Form Factor. The Modulator Test Cover Assembly shall be capable of attachment directly to the Modulator Housing. Transparent access covers shall provide visibility of the RF Section and safe access to the section shall be provided by suitable interlocks when the covers are opened. The Cover Assembly shall include a capacitor load which can be used to load the output of the Pulser when the Klystron Power Amplifier is disconnected while checking the Pulser. It shall also contain a capacitor divider for monitoring the output pulse of the Pulser and bring the divided signal to the test points on the f rent panel. A shelf and tie-down straps shall be provided to mount a multimeter. An accessory box with a hinged cover and latch shall be provided for storage of the High Voltage Probe (not supplied). The overall dimensions of the Cover shall be 31 -5/8 inches by 13-3/8 inches by 6-3/8 inches.

3..1.6.8.2.3 High-Voltage Probe. A high-voltage probe shall be provided for use in conjunction with the Modulator Test Cover Assembly to make direct diagnostic measurements up to 20, 000 Vdc on the Pulser and high-voltage power supplies of the radar. The probe characteristics are:

(a) The probe shall provide an attenuation of 100 to 1 of the input voltage.

(b) The probe shall be suitable for use with the AN/ PSM-4 or other multimeters on the 250 V stale.

(c) The probe shall be equipped with banana plug connectors and a closed ground lug.

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3.4.1.6.8.2.4 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362.

3.4. 1.6.8.3 Stalo Extension Kit.

3.4.1.6.8.3.1 Function. The Stalo Extension Kit shall permit operation of the RF 0scillator, when it is removed from the system on the table-rack and placed on the bench table top for maintenances

3.4.1.6.8.3.2 Form Factor. The Stalo Extension Kit shall consist of an electrical Patch Cable Assembly and an Air Adapter.

3.4.1.6.8.3.2.1 Patch Cable Assembly. The Patch Cable Assembly shall be a cable 36 ± 1 inches long. It shall connect the RF Oscillator to the systern via connectors 3J1 and 3P1, 3J2 and 3P2, 3J3 and 3P3, 3J4 and 3P4, and 3J5 and 3P5. These connectors shall be wired pin to pin with wire per MIL-W-16878 and RG-195/U subminiature coaxial cable per MIL-C-17/70 as required.

3.4.1.6.8.3.2.2 Air Adapter. The Air Adapter shall not exceed 65 ± 1 inches in length and shall furnish cooling air to the RF 0scillator, from the Plenum Chamber on the Pallet Bench.

3.4.1.6.8.4 Kit, Waveguide Shelf Extension .

3.4.1.6.8.4.1 Function. The Waveguide Shelf Extension Kit shall provide the means by which the Receiver Waveguide Assembly and the Transmitter Waveguide Assembly of the AN/APG-59 Radar Set and the components mounted thereon are made accessible for adjustment, servicing, and maintenance. The Waveguide Shelf Extension Kit is replaceable by the Waveguide Extension MX 849l/AWG-10 and in some locations the Radar Maintenance Table-Rack will be so configured. Some cables from the Waveguide Shelf Extension Kit are retained when replacing it with the Waveguide Extension.

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3.4.1.6.8.4.2 Form Factor. The Extension Kit shall be in the form of Posts, Microwave Shelf Extensions, Cable Assemblies and other accessories which shall permit the Radar Set Receiver and Transmitter Waveguide As - semblies to be lowered a nominal 10 inches and operated in the lowered position.

3.4.1.6.8.4.3 Contents. The Waveguide Shelf Extension Kit shall consist of the following items.

(a) Posts, Extension Tool
(b) Extensions, Microwave Shelf
(c) Extension Air Duct
(d) Cable Assemblies

3.4.1.6.8.4.3.1 Post Extension Tool.

3.4.1.6.8.4.3.1.1 Function. Four identical Post Extension Tools shall be furnished. The Post shall provide the means for supporting the Radar Set Receiving and Transmitting Waveguide Assemblies in the lowered position.

3.4.1.6.8.4.3.1.2 Form Factor. The Post Extension Tool shall be made from l/2-inch hexagonal bar stock and shall have an effective length of 10 inches.

3.4.1.6.8.4.3.1.3 Marking. Each Post shall be marked: USED WITH MICROWAVE SHELF (609R434) 4 REQ'D SHELF EXTENDER PART OF I LEVEL BENCH.

3.4.1.6.8.4.3.2 Extension, Microwave Shelf.

3.4.1.6.8.4.3.2.1 Function. Three identical Microwave Shelf Extensions shall be provided. Thes e Extensions shall provide wave guide continuity to the Radar Set Receiver and Transmitter Waveguide Assemblies when they are lowered for access.

3.4.1.6.8.4.3.2.2 Form Factor. The Microwave Shelf Extensions shall be constructed from nominal 1/2-inch by 1-inch wave guide. Each end of the ex- tens ions shall be fitted with wave guide flanges. Each extension shall be a nominal 10-inches long flange surface to flange surface.

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3.4.1.6.8.4.3.2.3 Marking. Each extension shall remarked: USED WITH MICROWAVE SHELF (609R434) 3 REQ'D SHELF EXTENSION PART OF TABLE RACK RADAR MAINTENANCE.

3.4.1.6.8.4.3.3 Extension, Air Duct.

3.4.1.6.8.4.3.3.1 Function. The Extension Air Duct shall permit cooling air to be supplied to the Para Amp Pump Housing of the AN/APG-59 Radar Set when the Receiver and Transmitter Wave guide Assemblies of the Radar Set are lowered for access.

3.4.1.6.8.4.3.3.2 Form Factor. The Extension Air Duct shall be fabricated from rectangular tubing a nominal 1 inch by 1/2 inch with an overall length of 11- 1/4 inches. The Air Duct shall be fitted with attachment fittings and shall be sealed at the ends. The Air Duct shall be offset and shall contain an air inlet and an air outlet, each approximately 5/8 inch in diameter arranged such that they are in the same plane effectively 10 inches apart.

3.4.1.6.8.4.3.3.3 Marking. The Extension Air Duct shall be marked: USED WITH MICROWAVE SHELF (609R434) AIR DUCT EXTENSION TOOL PART OF TABLE RACK RADAR MAINTENANCE.

3.4.1.6.8.4.3.3.4 Cable Assemblies. One each of the following Cable As - semblies shall be provided to permit normal operation of the AN/APG-59 Radar Set Receiver and Transmitter Wave guide Assemblies when they are lowered for access.

(a) Receiver Waveguide Assembly, Patch Cable Assembly No. I - This assembly shall consist of four individual cables 20 ± 1 inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J12 and 2A8P12; 2A8J13 and 2A8P13; 2A8J14 and 2A8P14; 2A8J27 and 2A8P27. All connectors shall be wired with RG-195/U subminiature coaxial cable per MIL-C-17/70.

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b) Receiver Waveguide Assembly, Patch Cable Assembly No. 2- This assembly consists of four individual cables 20 ± 1 inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J11 and 2A8P11; 2A8J15 and 2A8P15; 2A8J16 and 2A8P16; 2A8J28 and 2A8P28. All connectors shall be wired with RG-195/U subminiature coaxial cable per MIL-C-17/70.

(c) Receiver Waveguide Assembly, Patch Cable Assembly No. 3 - This assembly shall consist of three individual cables 16 ± 1 inches long. The individual cables shall be formed into one main assembly, leaving the individual cables free at both ends to mate with system connectors 2A8J17 and 2A8P17; 2A8J18 and 2A8P18; 2A8J19 and 2A8P19. All connectors shall be wired with MIL-W-16878 high-voltage wire.

(d) Transmitter Waveguide Assembly, Patch Cable Assembly - This assembly shall consist of a cable 20 ± 1 inch long. The Cable Assembly shall mate with system connectors 2A2J2 and 2A2P2. The connectors shall be wired pin to pin with wire per MIL-W-16878. Pins H and B shall be shield tie-points at each end of the cable.

3.4.1.6.8.5 Stalo Terminations. Two electrical dummy loads shall be provided to terminate the outputs (f_o and f_o + 30) of the Radio Frequency Oscillater (Stale) O-1330C/APG-59, whenever the unit is operated in an extended position or when either or both of the output ports are disconnected from their corresponding flex guide assemblies, The two terminations shall be identical to each other and shall meet the following requirements:

(a) Frequency Range: 8.5 to 10.4 GHz

(b) Maximum VSWR: 1.20 over the frequency range

(c) Average Power to the Load 250 mW (max)

(d) Peak Power to the Load: -250 mW (max) .

(e) Mechanical Interface: The terminations shall have choke flanges for 1\2-inch by 1-inch W/C with suitable captive hardware. They shall attach to the wave guide bandpass filter 3A1FL1 and/or the power monitor assembly 3A1Ml.

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(Assemblies 3A1FL1 and 3A1M1 have cover flanges similar to UG-135/U, except they have 8-32-2B tapped holes.)

3.4.1.6.8.6 DC Voltage Source. A variable dc voltage source shall be provided with the Table Rack, Radar Maintenance as an auxiliary item for uses as required during test and maintenance. It shall be identical to the item provided for Flightline Maintenance in the Cable Assembly Group OA-827A/AWM-22, as specified by MIL-T-81516 para 3.4.5.5.n.

3.4.1.6.8.7 Extender Bracket.

3.4.1.6.8.7.1 Function. An Extender Bracket shall be provided for extension and support of the Pitch and Roll Servo Assembly of the Velocity Signal Computer The assembly shall fasten to the bracket in such a manner that all resolvers are accessible for alignment adjustments. It shall be possible for one person to remove the assembly from the computer and mount it on the alignment bracket unaided. Disconnection or interruption of the resolver circuits at 15A1P2 for alignment purposes shall be possible.

3.4.1.6.8.7.2 Form Factor. The Extender Bracket shall consist of a bracket 12 inches by 9 inches by 6-318 inches, constructed such that the Pitch and R 011 Servo Assembly may be mounted on it. It shall be constructed such as to permit its being placed on the bench working surface in at least three positions and remain stable, in order to provide access to the servo alignment screws. Cabling shall be provided to permit extending the Pitch and Roll Servo Assembly 4 feet from' the Velocity Signal Computer while maintaining proper electrical integrity to perform all alignments.

3.4.1.6.8.8 Relay Tester.

3.4.1.6.8.8.1 Function. The relay tester shall provide testing for the standard eight-pin crystal can plug-in relays used in the AWG-10 and the current sensitive relays used in the high-voltage power supplies. The relay tester shall be able to test certain relays in their sockets. The relays to be tested shall include the following types:

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(a) Voltage Relays: 578R040H01 and H02
578R145H01, H02, H03, H07, H08, and H09
578 R147HOl and H02

(b) Current Relays: 578R040H11
578R147H1O

(c) Latching Relay: 576R148H03

(d) Reed Relay: 579R760H01

3.4.1.6.8.8.2 Power Requirements, The relay tester shall be capable of operating from a 50 to 400 Hz, 115 V source without any of the primary circuit breakers on the bench being closed. The relay tester shall be capable of being operated without being dependent upon the bench and shall have suitable circuit protection.

3.4.1.6.8.8.3 Coil Voltage. Voltage to the relay socket coil terminals shall be applied to binding posts which shall serve as test points for the coil voltage they shall enable the application of coil voltage to solder hook-type relays such as 478R145H05. Coil voltage of relays whose coil voltage is 26.5 V nominal shall be applied by the actuation of a momentary pushbutton switch. Switching shall be provided to control latching relay testing whereby it shall be possible to latch the relay and reset it.

3.4.1.6.8.8.4 Current Relay Coil Current. Provisions shall be made to test the current sensitive relays in socket 577R480H01. The test circuit shall vary the coil current to determine the specific current at which the relay becomes energized. Protection (fusing) shall be provided such that the relay under test or the tester shall no t be damaged during testing. A meter shall display coil current.

3.4.1.6.8.8.5 Overload Relays in Circuits. The tester shall provide testing for the following listed relays with their associated circuitry while the relays remain in their sockets. Power shall be supplied to the specified terminals in the following table while the current to the circuit is measured to an accuracy of ±5 percent.

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Test Circuit Connections	Allowable Operating Current	Test Circuit Voltage	Relay Monitor Points
2A5K2 E29 J1/1(gnd)	73 - 95 mA	+6 v	J1/4 & J1/8
2A5K2 E29 J1/5	436 -600 mA	+5. 5 v	J1/4 & J1/8
2A5K4 J1/1(gnd) E27	604 -838 mA	-5.5 v	J1/4 & J1/8
2A5K5 T2/3 J1/1(gnd)	1.6- 2.O	+24 V	J1/18&J1/17
2A6K1 J1/P J1/C	50 - 60 mA	+5. 5 v	J1/L & J1/8

When the operating cur rent between points listed in column 2 is within the limits specified in column 3, the circuits are functioning properly. Relay monitor points (for 3.4.1.6.8.8.6) are specified in column 5.

3.4.1.6.8.8.5.1 Overload Test Controls. Relay tester controls shall be capable of controlling current over the range required for each relay to be tested. Tester circuits shall be protected against damage by large in-rush currents by large capacitors connected acress several of the test circuit connections.

3.4.1.6.6.8.8.5.2 Harnesses. A harness shall be provided for the relays in the High-Voltage Power Supply and one for the relays in the C W Illuminator Power Supply. The harnesses shall be long enough to permit testing while the power supplies remain in the Modulator Housing with the electrical connectors disconnected.

3.4.1.6.8.8.6 Relay Contact Continuity. Proper relay contact continuity shall be determined by visual indication. Test current shall be 10 mA dc +0, -20 per cent. Open Circuit voltage shall not exceed 6 Vdc. (A lamp may not be used due to initial in-rush current.)

3.4.1.6.8.8.7 Controls. All operating controls shall be suitably marked so that an operator may operate the tester without reference to supporting documents (manuals or check lists). Controls for each of the tests and accept/ reject criteria for each test shall be evident.

3.4.1.6.8.9 Waveguide (W/G) Switch By pass. Provisions shall be made to disable and bypass the Waveguide Switch, (2A2S1) (mounted on the Transmitting Wave guide Assembly) in order to make it impossible to radiate from the antenna when the Radar Set is mounted on the bench in the shop.

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3.4.1.6.8.9.1 Waveguide Adaptor., The waveguide adaptor shall permanently connect the output of the Transmit Coupler (2A2DG2) to the input of the Dummy Load Coupler (2A2DC1). The wave guide adaptor shall be installed in place of the Wave guide Switch when the Radar Set is mounted on the bench. The Wave guide Switch, upon removal, may be stored in the bench for re-installation when the Radar Set is removed from the bench.

3.4.1.6.8.9.2 Bracket Assembly, A bracket assembly containing a dummy jack shall be provided. The jack shall mate with the Wave guide Switch connector 2A2S1P1. The dummy jack shall contain jumpers to bypass the +25 V enable signal normally interlocked by the Wave guide Switch. The dummy jack shall also contain provisions to feed the coil voltage for the Waveguide Switch to the RAD ANT indicator on the Aircraft Simulator panel (3.4.1.6.3.3.3.4) via the bench harness (3.4.1.6.10.1).

3.4.1.6.9 Fixture, Antenna Support.

3.4.1.6.9.1 Function. The Antenna Support Fixture shall mount the Anenna of the AN/APG-59 Radar Set which is mounted on the Radar Maintenance Table-Rack.

3.4..6.9.2 Form Factor. The overall dimensions of the Antenna Support Fixture shall be 45-3/4 inches by 34-5/8 inches by 75 inches with the guard ring folded and shall be 45-3/4 diameter by 55- 1/4 inches with the guard ring open. The weight of the fixture shall not exceed 200 pounds. The Antenna Support Fixture shall be similar in structure to the Antenna Maintenance Fixture, but shall not be supplied with the accessories provided with the Maintenance Fixture. The Antenna Support Fixture shall be designed for deck mounting either by bolting or welding, Two vertical

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side members shall extend upwards from a 22-1/4 inch by 24-5/8 inch rectangular base to position the Antenna Support which shall be composed of rotating cradle and mounting plate. The cradle, 39-.3 /4 inches above the deck, shall provide the fixture upon which the mounting plate rests. The mounting plate, in the horizontal position, shall hold the AN/APG-59 Antenna Base Assembly which shall be bolted to its surface. The mounting plate shall be capable of 360 degrees azimuth rotation and shall be capable of being locked securely in any desired position. The cradle shall be capable of being tilted 90 degrees from the horizontal plane by the use of a hand wheel and gear reducer. An aluminum guard rail, pivoted from an axis 14-1/2 inches above the mounting plate, shall encircle the antenna and shall act as a safety device to prevent accidental contact with antenna when it is operated in scanning mode. The guard rail shall be capable of being stowed out of the way in vertical position when the antenna is not scanning l An extend able oil pan shall be mounted below the cradle and between the vertical support members to retain hydraulic fluid leakage from the antenna connections. An interlock shall prevent antenna electrical operation any time the antenna cradle is rotated from its horizontal plane position.

3.4.1.6.9.3 Materials and Parts. Materials and parts of the Antenna Support Fixture Gear Reducer shall be of the best commercial quality, shall meet the performance and environmental requirements imposed by the equipment and shall be protected from corrosion such that its form and function shall not be impaired.

3.4.1.6.9.4 Associated Equipment The Hydraulic Supply is required to work in conjunction with the Antenna Support Fixture and shall be mounted on the base of the Fixture.

3.4.1.6.10 Interconnecting Test Harness.

3.4.1.6.10.1 Function. The Intereconnecting Test Harriess shall provide electrical intereconnection between the Table-Rack, the MCS and the System Harness. It shall provide or simulate that portion of the intereconnecting wiring normally provided by the F-4J Aircraft which is not within the Test

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and Control Rack. The harness shall provide for connecting the Wave guide Switch coil voltage from the dummy jack to the Test and Control Rack (3.4.1.6.8.9.2).

3.4.1.6.10.2 Form Factor. The Inter connecting Test Harness shall extend the length of the Table-Rack and shall interconnect the line Replaceable Unit (LRU's) of the AN/APG-59 Radar Set located throughout the several units of the table-rack listed in 3.4. 1 except those LRU's which are normally installed on the Barrel Assembly of the AN/APG-59 Radar Set. The Inter-connecting Cable Harness shall connect directly to the System Cable Harness at the Interconnecting Test Harriess Shelf to provide an interface with the. AN/APG-59 Barren Assembly LRU'S installed on the RF Section Support Assembly and the Reciever Section Bench Assembly.

3.4.1.6.10.3 Connectors. Connections to the AN/APG-59 Radar Set LRU's shall be made with approved connectors:

Reference Designation	Function
8P1	Power Supply (PP-4451/APG-59)
8P2*[8]	Power Supply (PP-4451/APG-59)
9P1*6	Command Indicator
9P2	Command Indicator
10P1*[2]	Radar Indicator
1OP2	Radar Indicator
llPl*[6]	Indicator Control
11P2*[6]	Indicator Control
11P3	Indicator Control
11P4	Indicator Control
15P1	Velocity Signal Computer
15P2	Velocity Signal Computer
15P3	Velocity Signal Computer

*2 See paragraph 6.5.2
*6 See paragraph 6.5.6
*8 See paragraph 6.5.8

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Reference Designation	Function
16P1	Launch Signal Computer
16P2	Launch Signal Computer
16P3*6	Launch Signal Computer
17P1*8	Pilot Command Signal Computer
17P2	Pilot Command Signal Computer
17P3	Pilot Command Signal Computer

Connection to the System Harness shall be made through the following connectors using the same pin designators as the corresponding connectors of the AN/APG-59.

Reference Designation	Function
7 J1 *6, *7, *8	Radar Main
7J2*6 & *8	Radar Main
7J3	BIT Main
7, J4*8	BIT Main
7J6	Simulated Doppler Main

3.4.1.6.11 System Harness.

3.4.1.6.11.1 Function. The System Harness shall replace the Rack, Electrical Equipment (LRU7) wiring harness which interconnects the AN/APG-59 Radar Set LRU's installed on the Barrel Assembly, and shall provide electrical interconnection between these LRU's, the table -rack and the Inter - connecting Test Harness*

3.4.1.6.11.2 Form Factor. The System Cable Harriess shall extend throughout the R. F. Section Support Assembly and the Receiver Section Bench Assembly to interconnect the AN/APG-59 Radar Set Barrel Assembly LRU's. The System Harriess shall connect directly to the Interconnecting Cable Harness at the Interconnecting Test Harness Shelf to form an interface with the table-rack and the remaining AN/APG-59 LRU's installed thereon.

*6 See paragraph 6. 5.6
*7 See paragraph 6. 5.7
*8 See paragraph 6. 5.8

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3.4.1.6.11.3 Connectors. Connections to the AN/APG-59 Radar Set Barrel Assembly LRU's shall be made with the following connectors:

Reference Designation	Function
1A3P1	Antenna
1A3P2*[6]	Antenna
1A3P3*[7]	Antenna
2A2P1*[5] & *8	Waveguide Assembly, Transmitting
2A2P12	Waveguide Assembly, Transmitting
2A2P13	Wave guide Assembly, Transmitting
2A2P14	Wave guide Assembly, Transmitting
2A2P18	Waveguide Assembly, Transmitting
2A2P42*[5]	Waveguide Assembly, Transmitting
2A2A1P2	Wave guide Assembly, Transmitting
2A2A5P1	Waveguide Assembly, Transmitting
2A3P1	Control-Filter, Power
2A3P2*[5]	Control-Filter, Power
2A4P1	Relay Assembly
2 A 4 P 2*[5&6]	Relay Assembly
2A8P11	Waveguide Assembly, Receiving
2A8P12	Wave guide Assembly, Receiving
2A8P13	Waveguide Assembly, Receiving
2A8P14	Waveguide Assembly, Receiving
2A8P15	Waveguide Assembly, Receiving
2A8P16	Wave guide Assembly, Receiving
2A8P27	Waveguide Assembly, Receiving
2A8P28	Waveguide Assembly, Receiving
3P1	0scillator, Radio Frequency
3P2	0scillator, Radio Frequency

*5 See Paragraph 6. 5.5
*6 See Paragraph 6. 5.6
*7 See Paragraph 6. 5.7
*8 See Paragraph 6. 5.8

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Reference Designation	Function
3 P3	Oscillator, Radio Frequency
3 P4	Oscillator, Radio Frequency
3 P5	Oscillator, Radio Frequency
4A1P1	Amp Assembly, Inter Frequency
4A1A1J1	Amp Assembly, Inter Frequency
4A1A2J1	Amp Assembly, Inter Frequency
4A1A3J1 *6	Amp Assembly, Inter Frequency
4A1A3J2	Amp Assembly, Inter Frequency
4A1A4J1	Amp Assembly, Inter Frequency
4A1A4J2	Amp Assembly, Inter Frequency
4A1A5J1	Amp Assembly, Inter Frequency
4A1A6J1	Amp Assembly, Inter Frequency
4A1A6J2	Amp Assembly, Inter Frequency
4A1A7J1	Amp Assembly, Inter Frequency
4A1A7J2	Amp Assembly, Inter Frequency
4A1A8J1	Amp Assembly, Inter Frequency
4A1A9J1	Amp Assembly, Inter Frequency
4A2P1	Analyzer, Doppler Spectrum
4A2P4	Analyzer, Doppler Spectrum
4A2P5	Analyze r, Doppler Spectrum
4A2A14Pl	Analyze r, Doppler Spectrum
4A3 P 1	Tracker, Target Data
4A3P2	Tracker, Target Data
4A3P3	Tracker, Target Data
5 A 1 P 1 *8&*2	Tracker, Range-Terrain Clearance
5AlP2 *6&*2	Tracker, Range-Terrain Clearance

*2 See Paragraph 6. 5.2
*6 See Paragraph 6. 5.6
*8 See Paragraph 6. 5.8

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Reference Designation	Function
5A1P3*7&*2	Tracker, Range-Terrain Clearance
5AlP4 *4, *6, & *2	Tracker, Range-Terrain Clearance
5A1P5*2	Tracker, Range-Terrain Clearance
5A2P1 *4, *6, & *8	Timer -Antenna Servo
5A2P2 *3	Timer-Antenna Servo
5A2P3*6	Timer-Antenna Servo
5A2P4	Timer-Antenna Servo
5A2P5	Timer-Antenna Servo
5A2P6	Timer-Antenna Servo
5A3P 1	Power Supply, Regulators
5A3P3	Power Supply, Regulators
6A1P1	Generator, Scan Pattern
6A1P2*8	Generator, Scan Pattern
6A2P1*2 & *6	Power Supply-Osc. and CW Offset
6A2P2*2	Power Supply-Osc. and C W Offset
6A2P3 *2 & *6	Power Supply-Osc. and CW Offset
6A2P4*2	Power Supply-Osc. and CW Offset
6A2P5*2	Power Supply-Osc. and CW Offset
6A2P6*2	Power Supply-Osc. and CW Offset
7P5	Coaxial Attenuator
7J7	Adapter
7P7	Adapter

*2 See Paragraph 6. 5.2
*3 See Paragraph 6. 5.3
*4 See-paragraph 6. 5.4
*6 See Paragrpah 6. 5.6
*7 See Paragraph 6. 5.7
*8 See Paragraph 6. 5.8

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Connection to the Interconnecting Test Harness shall be made through the following connectors using the same pin designation as the corresponding connectors of the AN/APG-59.

7 P 1 *5, *6, *7, & *8	Radar Main
7P2 *6 & *8	Radar Main
7P3	BIT Main
7 P 4 *8	BIT Main
7P6	Simulated Doppler Main

3.4.1.6.12 Test Set Interface. The Table Rack, Radar Maintenance shall provide any necessary interface with other items of the Test Set described herein. The interface shall include mechanical mounting provisions, electrical connections for power, and test signals as necessary to the extent required for efficient maintenance routines. Provisions shall be made for, but not limited to, the following items:

(a) Antenna Maintenance Fixture
(b) Wave guide Extension
(c) Test Set, Missile control System
(d) Analyzer, RF Noise

3.4.2 Fixture, Antenna Maintenance MX-8359B/AWM-21.

3.4.2.1 Function. The equipment shall provide a separate facility to service the AN/APG-59 antenna. It shall permit antenna maintenance without seriously interfering with work on the Table - Rack.

*5 See paragraph 6. 5.5
*6 See paragraph 6. 5.6
*7 See paragraph 6. 5.7
*8 See paragraph 6. 5.8

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3.4.2.2 Form Factor. The Antenna Maintenance Fixture shall be deck mounted on a 22- 1/4 inches by 24-5/8 inches rectangular base. The overall dimensions shall be 45-3/4 inches by 24-5/8 inches by 75 inches with the guard ring folded and shall be 45-3/4 inches by 55-1/4 inches with the guard ring open. The total weight of the equipment including cables shall not exceed 450 pounds.

3.4.2.3 Materials and Parts. Materials and parts of the gear reducer, and hand wheel of the Antenna Support Fixture shall be of the best commerical quality, shall meet the performance and environmental requirements imposed by the equipment and shall be protected from corrosion such that its form and function shall not be impaired.

3.4.2.4 MTBF. When maintained in accordance. with applicable preventive maintenance procedures, the equipment shall demonstrate at least 6,000 hours MTBF. Accept-reject criteria shall be those defined by paragraph 4.2.8.1 of MIL-STD-781.

3.4.2.5 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H- 15362,

3.4.2.6 Primary Input Power Requirements. The Antenna Maintenance Fixture shall obtain all input power from the Table Rack, Radar Maintenance FN-212D/AWG-10 through interconnection cabling described herein and shall require no external power

3.4.2.7 Contents of Unit. The Antenna Maintenance Fixture shall consist of the following items:

(a) Antenna Support Fixture
(b) Antenna Control Box
(c) Hose Assembly, Hydraulic
(d) Cable Assemblies, Electrical
(e) Hydraulic Panel

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3.4.2.7.1 Antenna Support Fixture.

3.4.2.7.1. 1 Function. The Antenna Support Fixture shall physically support the AN/APG-59 Antenna for servicing. The fixture shall also incorporate provisions for holding the Antenna Control Box.

3.4.2.7.1.2 Form Factor. The Antenna Support Fixture shall be designed for deck mounting either by bolting or welding. Two vertical side members shall extend upwards from a 22- 1/4 inches by 24-5/8 inches rectangular base to position the antenna support which shall be composed of a rotating cradle and a mounting plate. A cradle, 39-3/4 inches above the deck, shall provide the fixture upon which a mounting plate rests. The mounting plate, in the horizontal position, shall hold the AN/APG-59 Antenna Base Assembly which shall be bolted to its surface. The mounting plate shall be capable of 360 degree azimuthal rotation and shall permit locking securely in any desired position. The cradle shall be capable of being tilted 90 degrees from the horizontal plane by use of a hand wheel and gear reducer. An aluminum guard rail, pivoted from an axis 14-1/2 inches above the mounting plate, shall encircle the antenna and shall act as a safety device to prevent accidental contact with the antenna when it is operated in a scanning mode. The guard rail shall be capable of being stowed out of the way in vertical position when the antenna is not scanning, An expendable oil pan shall be mounted below the cradle and between the vertical support members to retain hydraulic fluid leakage from the antenna connections. An interlock shall prevent antenna electrical operation any time the antenna cradle is rotated from its horizontal plane positions

3.4.2.7.2 Antenna Control Box.

3.4.2.7.2.1 Function. The Antenna Control Box shall provide the means to allow the operator to check the electromechanical functions associated with the AN/APG-59 antenna. The unit shall allow the antenna to be positioned either by manual contro1 or by automatic scan.

3.4.2.7.2.2 Form Factor, The Antenna Control Box shall be an enclosed control panel with dimensions of 12 inches by 11-1/2 inches by 7-1/2 inches

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with a 1- 1/4 inch mounting flange along one end which shall provide switches, controls, test points and connectors as required to perform the function described in section 3.4.2.1. It shall be bracket mounted and accessible from the front of the Antenna Support Fixture.

3.4.2.7.2.3 Inputs. Inputs to the Antenna Control Box shall be as follows:

(a) AC power in accordance with 3.4.2.6.

(b) DC power in accordance with 3.4.2.6.

(c) AN/APG-59 Antenna Azimuth Potentiometer output: O.346 ± 11 per-cent Vat/degree off bore sight.

(d) AN/APG-59 Antenna Elevation Potentiometer output: O.346 ± 11 per-cent Vat/degree off boresight.

(e) Antenna Position Indicator (API) outputs: Four signal leads with either O to 1.2 Vdc or 11 to 13 Vdc.

3.4.2.7.2.4 Outputs. Outputs from the Antenna Control Box shall be as follows :

(a) AN/APG-59 Antenna Azimuth Valve Command Signal: 2.19 ± 20 percent milliamperes (d) per degree of position error.

(b) AN/APG-59 Antenna Elevation Valve Command Signal: 1.37 ± 20 percent mA per degree of position error.

(c) AN/APG-59 Antenna Azimuth and Elevation Induction Potentiometer Excitation: Phase A, 26.4 Vac ± 8 percent, 400 Hz.

(d) AN/APG-59 Antenna Feedhorn Spin Motor Excitation: 3 -phase, 115/200 Vac *5 percent, 400 Hz.

(e) Simulated Acquisition: 27 ± 2 Vdc

(f) Monopulse Mode Select: 27 ± 2 Vdc

(g) 12 Vdc B+ Supply: 12 ± 0.6 Vdc

(h) BIT Control Number 1: 27 ± 2 Vdc

(i) BIT Control Number 2: 27 ± 2 Vdc

(j.) BIT Control Number 3: 27 ± 2 Vdc

(k) BIT Control Number 4: 27 ± 2 Vdc

{1) BIT Control Number 5: 27 ± 2 Vdc

(m) Spoiler Control: 27 ± 2 Vdc

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3.4.2.7.2.5 Controls. The following controls shall be provided on the front of the Antenna Control Box. An associated indicator light shall be provided for section 3.4.2.7.2.5 (d).

(a) SCAN MOTOR POWER - A two-position, ON-OFF toggle switch shall be provided on the panel for the purpose of applying 11 5/200 Vat, 400 Hz, 3-phase power to the scan drive motor in the Antenna Contro1 Box.

(b) AUTO SCAN DIRECTION (Automatic Scan Direction) - A three-position switch shall be 'provided for the purpose of programming three separate scan commands into the AN/APG-59 Antenna. Positions shall be marked AZ, EL, and 45°. The commanded scan coverage shall be 110 degree + 10 degree -O degree respectively in the azimuth plane, 110 degree + 10 degree, -O degrees in the elevation plane and a simulated roll of 45 degree which shall drive the Antenna in azimuth and elevation simultaneously.

(c) SPIN MOTOR POWER - A two-position, ON-OFF toggle switch shall be provided to apply 115/200 Vat, 400 Hz, 3-phase power to the AN/APG-59 Antenna Feedhorn Motor.

(d) HYDRAULICS - A two-position momentary ON-OFF toggle switch shall be provided for contro1 of the Hydraulic Supply to furnish hydraulic power to the AN/APG-59 Antenna. A green lamp shall be located beside the toggle switch to indicate that hydraulic power has been applied to the antenna.

(e) SCAN CMD (Scan Command Select) - A two-position AUTO, MAN (Manual) toggle switch shall be provided. The AUTO position shall provide a scan rate of 60 ± 6 degrees per second. The MAN position shall provide the facility to position the antenna by use of the AZIMUTH MANUAL COMMAND and ELEVATION MANUAL COMMAND controls.

(f) AZIMUTH MANUAL COMMAND - A potentiometer r shall be provided to position the AN/APG-59 Antenna to any static azimuth position desired. This potentiometer shall provide an input to the azimuth scan-command - servo amplifier when selected by the SCAN CMD switch.

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(g) ELEVATION MANUAL COMMAND - A potentiometer shall be provided to position the AN/APG-59 Antenna to any static elevation position desired. This potentiometer r shall provide input to the elevation scan-command Servo-Amplifier when selected by SCAN CMD switch.

(h) WG/SP TEST (Waveguide/Spoiler Test) - A three-position toggle switch shall be provided to apply +28 Vdc to either the AN/APG-59 Antenna Waveguide Switch or the Spoiler. Positions shall be marked WAVEGUIDE, OFF, and SPOILER.

(i) SPIN MOTOR - A two-position toggle switch shall be provided to control the spin motor drive for the AN/APG -59 Antenna Feedhorn. The positions shall be labeled SPIN and STOP.

(j) API +28 VDC BIT (Antenna Position Indicator, BIT Relay) - A five -position switch shall be used to provide the facility for energizing the BIT relays on the AN/APG-59 Antenna and to provide the means for B+ excitation to the API photo sensors. Positions shall be marked OFF, 1, 2, 3, and 4.

3.4.2.7.2.6 Protective Devices. The following protective devices shall be provided on the front of the Antenna Control Box:

(a) Circuit Breakers - Two circuit breakers shall be provided to protect the primary power source from overloading due to electrical malfunction on the Antenna Maintenance Fixture.

. (1) +28 Vdc - 5A

(2) 400 Hz - 115/200 Vat, 3 phase, 7-1/2A

3.4.2.7.2.7 Electrical Connectors. Three connectors shall be provided on the control panel for interconnection between the AN/APG-59 Antenna and the Table-Rack.

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3.4.2.7.2.8 Test Points. Five test points (API OUTPUTS) shall be provided on the control panel to monitor the AN/APG-59 Antenna outputs. Test points shall be marked A, B, C, D, and GRD.

3.4.2.7.2.9 Servo Amplifier. The Antenna Control Box shall contain a Servo Amplifier for the azimuth channel and one for the elevation channel. The amplifiers shall accept two inputs:

(a) Antenna inputs as specified in 3.4.2.7.2.3 (c) and (d)

(b) SCAN CMD of either AUTO or MANUAL as selected (see section 3.4.2.7.2.5 (e)) and provide an output as specified in sections 3.4.2.7.2.4 (a) and (b).

3.4.2.7.3 Hose Assembly Hydraulic. Two hydraulic hose assemblies shall be 20 feet in length, consisting of insulated hydraulic hoses to connect the AN/APG-59 Antenna on the Antenna Maintenance Fixture to the Hydraulic Panel. Two additional hose assemblies shall be provided to connect the Hydraulic Pumping Unit to the Hydraulic Panel when this panel is installed on the Antenna Support Fixture. The hoses shall be medium pressure with teflon lining and corrosion-resistant braid reinforcement. The Assembly shall conform to MIL-H-25579, MIL-H-27267 and MIL-F-27272. The hoses shall have end fittings to mate with standard MS flareless tube per MS-33514.

3.4.2.7.4 Cable Assemblies, Electrical.

3.4.2.7.4.1 Function. The Electrical Cable Assemblies shall provide the patch cables necessary to operate the specified electrical function of the Antenna Maintenance Fixture.

3.4.2.7.4.2 Cable Assembly, Primary Power and Hydraulic Interlock, 64W6. The Primary Power and Hydraulic Interlock Cable shall be nominally 51.5 feet in length, and it shall provide primary power to the Antenna Control Box from the sidebench and it shall provide safety interlocking with the Hydraulic Panel so that hydraulic power cannot inadvertently be applied to the Antenna Maintenance Fixture when the bench antenna is selected.

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3.4.2.7.4.3 Cable Assembly, Antenna Interconnecting, 64W1. The Antenna Interconnecting Cable Assembly shall be nominally 6 feet in length including the connectors and the two-branch division. It shall provide electrical inter connection between the Antenna Control Box and the AN/APG-59 Antenna on the Antenna Maintenance Fixture.

3.4.2.7.5 Hydraulics Panel.

3.4.2.7.5.1 Function. This unit shall permit the Hydraulic Supply to deliver fluid to either the .Antenna mounted on the Antenna Support Fixture or the Antenna Maintenance Fixture. An electrical interlock shall also be provided to assure control of the Hydraulic Supply by the facility using it.

3.4.2.7.5.2 Form Factor. The Hydraulic Panel shall consist of the hydraulic transfer valve, fittings, and check valves on a support bracket. This panel, although a part of the Antenna Maintenance Fixture, shall attach to the Antenna Support Fixture. The overall dimensions, exclusive of the interconnecting lines, shall be 15 7/8 inches by 71/8 inches by 14 1/2 inches. The weight of the unit, excluding interconnecting lines, shall not exceed 27 pounds.

3.4.2.7.5.3 Fittings. The Panel shall have the following quick disconnect fittings:

Function
Connects pressure side of Hydraulic Supply to Panel
Connects return side of Hydraulic Supply to Panel
Connects pressure side of Panel to Antenna Support Fixture antenna
Connects return side of Panel to Antenna Support Fixture antenna
Connect pressure side of Panel to Antenna Maintenance Fixture antenna
Connects return side of Panel to antenna Maintenance Fixture antenna

3.4.2.8 Associated Equipment. The Hydraulic Pumping Unit is required to work in conjunction with the Antenna Maintenance Fixture.

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3.4.3 Waveguide Extension (WE) MX-849l/AWG-10.

3.4.3.1 Function. The WE shall be designed to mount on the AN/APG-59 Radar Set structure in place of the Receiving Wave guide Assembly. It shall provide for mounting and support of the Receiving Waveguide Assembly in a position approximately 10 inches below and 6 inches forward of its original location. Provision shall also be made to mount and support a Bit Target Generator (BTG), Switching, control, and monitoring functions shall be incorporated to provide capabilities for alignment of the Parametric Amplifiers, using a. sweep generator; calibration of the BIT Target Generator; and calibration of the 17 -dB variable attenuator. The se capabilitiess shall be provided by incorporation of the following functions:

(a) The Radio Frequency Oscillator Transmitted Frequency shall be provided to the Wave guide Shelf in any of the three operating modes (Pulse, Pulse Doppler and Continous Wave).

(b) Switching the Main Receiver and Auxiliary Receive r Parametric Amplifier inputs from the Radar Receiving Wave guide outputs of the AN/APG-59 Radar Set to the output of the X-Band Sweep Generator.

(c) Insertion of a 19 -dB fixed attenuator between the Sweep Generator output and the Parametric Amplifier inputs.

(d) Insertion of a 17 -dB fixed attenuator on the input side of the Auxiliary Receiver Parametric Amplifier input.

(e) Calibration of either BTG (Radar or Test) in the three signal modes (Short, Long, and Pulse Doppler).

(f) Equalization of the BTG Reference Signal path with the BTG Calibration path.

(g) Diverting the input signal from the BTG under test and applying the signal to the radar BTG. Only one BTG shall receive heater power or BIT Doppler Controller (BDC) at any one time.

(h) Comparison of the BTG output signal with the BTG input signal. The input signal shall be suitably attenuated for comparison purposes.

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(i) Monitoring of the Test BTG crystal currents.

(j) Switching the oscilloscope video and sync inputs between the Pulse Doppler Mode and Pulse Mode.

(k) X-band frequencies used by the AN/APG-59 Radar Set from the Sweep Generator shall be measured to an accuracy of ± 0.1% of the measured frequency.

(1) Turning the WE on and off from the Test and Control Rack of the Radar Maintenance Table -Rack.

3.4.3.2 Form Factor. The configuration of the WE shall be such that it does not extend beyond the contour of the Radar Maintenance Table-Rack. It shall be compatible with both the Centerline and Side Bench configurations of the Radar Maintenance Table - Rack without modification, shall permit maintenance operations on the Radar Set without the necessity for disassembly. Electrical interfacing shall contain no permanent (solder) type connections and shall utilize quick- disconnect type connections wherever possible. Size of the WE shall not exceed 18 x 18 x 30 inches. Weight of the WE exclusive of system LRU'S, shall not exceed sixty (60) pounds.

3.4.3.3 Interface. Performance of the Wave guide Extension is dependent upon integration with the Test Set, Missile Control System, AN/AWG-10 Side Bench or Centerline Bench. "Timing and BDC signals, power, X-band energy, and cooling air are obtained by interfacing with the Bench Radar System, and the Side Bench or Certerline Bench Test- Rack. The Wave guide Extension (WE) shall be designed as a part of the Intermediate Level Facility and shall be compatible with both Centerline and Sidebench configuration of the Radar Maintenance Table -Rack. This combination shall be for test purposes and the electrical interface shall contain no permanent (solder) type connections. Quick disconnect type connections shall be used wherever possible.

3.4.3.4 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362.

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3.4.3.5 Contents of Unit. The WE shall consist of the following items:

(a) Microwave Structure

(b) Microwave Control Panel Assembly

(c) Interfacing Cabling

(d) Installation Kit

3.4.3.5. 1 Microwave Structure.

3.4.3.5. 1.1 Function. The basic Microwave Structure shall provide the physical and RF energy interfaces between the AN/APG-59 Radar Set and the Receiving Microwave Assembly and/or the Bit Target Generator. It shall include the microwave components required to perform the test and/or calibration functions. The Microwave Structure wiring harriess shall provide for interconnection between the Microwave Control Panel Assembly, the microwave components, and the AN/APG-59 Radar Set. Design of the harness shall conform to MIL-H- 15362 and shall be color coded in conformity with MIL-STD-681 where applicable.

3.4. 3.5.1.2 Form Factor. The basic structure shall be a U- shaped fixture with provision to fasten the WE to the AN/APG-59 Radar Set structure in the space normally occupied by the Receiving Microwave Assembly. Microwave components; such as waveguide switches, coaxial switches, fixed and variable attenuators, hybrids, isolators, directional couplers, frequency meters, etc; shall be included in configuration to permit interfacing between the Radar Set and the Receiving Microwave Assembly and connection to the Bit Target Generator. Various circuit configurations, as required for test and/or calibration shall be switched electrically and require no physical relocations of microwave components. Provision shall be made to support the Receiving Microwave Assembly and Bit Target Generator, and to provide for ducting of cooling air from the Radar Set structure to the Receiving Microwave Assembly.

3.4.3.5.2 Microwave Control Panel Assembly.

3.4.3.5.2.1 Function. Except for WE turn on and off, the Microwave Control Panel Assembly shall provide all s wit thing and control functions

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required for the operation of the Waveguide Extension, and shall be composed of the following:

(a) Microwave Control Panel

(b) Diode Gate Driver

3.4.3.5.2.2 Form Factor. The Microwave Control Panel Assembly shall be approximately 5.25 x 9 x 9 inches, exclusive of switches and receptacles, and shall be cantilevered off the left side of the Wave guide Extension on a hinged support, to provide access for removal of the RFO for servicing.

3.4.3.5.2.3 Microwave Control Panel. The Microwave Control Panel shall contain the following control switches, indicators, and test jacks:

(a) MODE SELECT Switch - The MODE SELECT Switch shall provide for selection of any of the following modes:

(1) O, 40 and 79 positions provide for interjection into the shelf of STALO Center Frequency in the CW Mode. When used in conjunction with other controls, Pulse or Pulse Doppler Modulation shall be added. Attenuators shall provide up to 80 dB of attenuation for this signal.

(2) PA RAMP: Switching RF energy supplied by the Bench X-Band Sweep Generator to the main or Auxiliary Receiver channels in the Receiving Waveguide Assembly.

(3) SHORT, LONG or PD: Energizing the BTG in either of the three signal modes.

(4) BENCH: Select the BTG bypass signal path in one signal mode so that it may be equalized to the normal signal path with BTG shorted.

(5) XMIT BY-PASS - SHORT or PD positions: Switch RF, heater power, and BDC from the BTG under test to the radar BTG in either SHORT or PD positions,

(b) REF (Reference Enable) Switch - The REF switch shall cause the center frequency to be modulated when the Mode Select is in the O, 40 or 79 position. When the Mode Select is in the SHORT, LONG or PD position, it shall select the BTG bypass path and the proper attenuation.

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(c) 19 dB Switch- The 19 dB switch shall interpose a fixed attenuation of 19 dB between the X-Band Sweep Generator and the Receiver inputs when the Mode Select is in the PARAMP position.

(d) 17 dB Switch - The 17 dB switch shall interpose a fixed attenuator of 17 dB into the input path of the Auxiliary Receiver channel when the Mode Select is in the PARAMP position.

(e) VIDEO SYNC Switch - The VIDEO SYNC shall allow the selection of either the DSA or the B-Gun Video oscilloscope presentations. When the Mode Select is in position 0, 40 or 79 and the REF Switch is activated, it allows the operator to choose between Pulse and Pulse Doppler Modulation.

3.4.3.5.2.3.1 Form Factor, The Microwave Control Panel shall be approximately 5.25 x 9 inches.

3.4.3.5.2.4 Diode Gate Drive r

3.4.3.5.2.4.1 Function. The Diode Gate Drive r shall furnish Diode Gate and Stalo Gate control signals fox the three modes of operation of the WE.

3.4.3.5.2.4.2 Form Factor. The Diode Gate Driver shall consist of a plug -in printed wiring component board approximately 4.3 x 6 inches.

3.4.3.5.3 Interfacing Cables

3.4.3.5.3.1 Function. The interfacing cables shall provide those inter - connections between the LRU's of the AN/APG-59 Radar set and the WE that are not provided for by the Microwave Structure wiring harriess.

3.4.3.5.3.2 Form Factor. The form factor of the Interfacing Cables shall be such that all necessary interconnections are made without permanent (solder) type connections. Interfacing connectors shall be compatible with the mating connectors of the AN/APG-59 and the Centerline or Side Benches. These cables shall conform to MIL-H-15362 and shall be color coded in conformance with MIL-STD- 681 where applicable.

3.4.3.5.4 Installation Kit.

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3.4.3.5.4.1 Function. The Installation Kit shall contain all components, wire, hardware, and other materials required to complete the installation and interconnections of the WE. When installed, the kit shall complete the modifications of the Centerline or Side Benches to provide the following:

(a) WE Turn on and Off from the MCS Control Panel of the Test Control Rack

(b) Transmitter on Pulse connection

(C) B-Gun Video Line Connection

(d) DSA Output Line connection

(e) DSA Start Line connection

3.4.4 Cable and Adapter Group, 0A-8421B/AWM-21. A Cable and Adapter Group shall be supplied as specified in MIL-T-81517(AS).

3.4.5 Tool Kit, Test Set, TK-217/AWM-21. A Test Set Tool Kit shall be supplied as specified in MIL-T-81517(AS).

3.4.6 Tool Kit, Maintenance TK-216/AWM-21. A Maintenance Tool Kit shall be supplied as specified in MIL-T81517(AS).

3.4,7 Fixture, Antenna Transporting MX-7339/AWM-21. An Antenna Transporting Fixture shall be supplied as specified in MIL-T-81517(AS).

3.4.8 Test Set, Reference Rate Gyro, TS-2512/AWM-21. A Reference Rate Gyro Test Set shall be supplied as specified in MIL-T-81517(AS).

3.4.9 Generator, Signal SG-696B/AWM-21. A Signal Generator shall be supplied as specified in MIL-T-81517(AS).

3.4.10 Compressor-Dehydrator, Electronic Equipment, HD-755/AWM-21. A Compressor-Dehydrator shall be supplied as specified in MIL-T-81517(AS)

3.4.11 Test Set, Missile Control System, TS251lC/AWG-10. A Test Set Missile Control System shall be supplied as specified in MIL-T-81741(AS).

3.4.12 Counter, Electronic, Digital, CP-870/AWM-21. An Electronic Counter shall be supplied in accordance with MIL-C.-24165.

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3.4.13 Oscilloscope OS-180/AWM-21. An Oscilloscope shall be supplied as specified in MIL-T-81517AS.

3.4.14 Analyzer, Radio Frequency Noise TS-2573E/AWG-10. A Radio Frequency Noise Analyzer shall be supplied. The designation of the military s pacification is yet to be determined.

3.4.15 Cabinet, Electronic Equipment - CY-6280B/AWM-21. An Electronic Equipment Cabinet as specified in MIL-C-81583 shall be supplied to contain the TS-2573E/AWG-10.

3.4.16 Signal Generator, SG-844/AWM-21. A Signal Generator as specified in MIL-T-81517(AS) shall be supplied,

3.4.17 Voltmeter, Digital ME-371/AWM-21. A Digital Voltmeter as specified in MIL-T-81517(AS) shall be supplied.

3.4.18 Radar Set, AN/APG-590 A Radar Set shall be provided in accordance with MIL-R-81114(AS).

3.4.19 Multimeter, TS-505/U. A Multimeter TS-505/U as specified in MIL-M-55088 is required for use with the AWG-10 Side Bench Test Set. This Multimeter is General and Standard equipment and is not supplied as part of the AWG-10 Side Bench Test Set.

3.4.20 Multimeter, AN/PSM-4. A Multimeter AN/PSM-4 as specified in MIL-M-17096 is required for use with the AWG-10 Side Bench Test Set. This Multimeter is general and standard equipment and is not supplied as part of the AWG-10 Side Bench Test Set.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(a) Preproduction Tests

(b) Acceptance Tests

(c) Life Tests

NOTE: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government, The Government reserves the right to perform any inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests. Preproduction tests shall be made on one or more. equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(a) Contractor's Demonstration Tests

(b) Service Approval Tests

4.2.1 Contractor's Demonstration Tests. Contractor's demonstration tests shall be accomplished uncle r the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of section 4.5 The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service for a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests. At the completion of the contractor's demonstration test and when requested by the procuring activity the equipment shall be delivered to the Government for additional testing. This additional testing may consist of duplicating tests previously conducted and other test necessary to determine compliance with all applicable design and performance requirements.

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4.2.3 Scope of Tests. Preproduction tests shall include all tests necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests include environmental testing in accordance with procedures of MIL-T-5422 with Test Limits modified to correspond with the limits specified herein or as specified in the detail equipment specification. Interference tests shall be conducted in accordance with MIL-I-6181 for those items designed in accordance with MIL-T-21200.

4.2.4 Preproduction Approval, Approval of the preproduction models shall be by the Naval Air Systems Command upon satisfactory completion of all tests. All preproduction models, except those consumed in testing or those rendered irreparable for their intended use, shall be returned to the Contractor for his use in the f abdication and testing of equipment to be submitted for acceptance. The preproduction models shall not be considered as equipments under this contract, however, such models may be reworked and submitted for acceptance as production equipments. Incorporation of design changes, in delivered production equipments, found to be necessary as a result of the preproduction test program, shall be the subject of separate and independent negotiations on an individual basis.

4.3 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Special Tests

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4.3.1 Individual Tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Tests

(c) Reliability Test

4.3.1.1 Examination of Product. Each equipment shall be examined care - fully to determine that the material, workmanship, and interchangeability requirements have been met.

4.3.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test. Each equipment shall be operated for a period of 6 hours without failure under the conditions specified herein. Only those adjustments will be permitted which can be made by using such controls and adjustments as are accessible to the operator during the normal use of the equipment. The equipment is subjected to the following environmental tests:

(a) Temperature	Room Ambient
(b) Humidity	Room Ambient .
(c) Vibration (10 minutes) .	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum double amplitude of 0. 036 inch.

The equipment shall be vibrated (without vibration isolators between the equipment and test fixtures) for a period of 10 minutes prior to the begin- ning of the 6-hour period of operation. Where fessible, the equipment shall

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be operated during these vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactor y performance is, not necessarily required during vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hours period of operation following the 10-minute vibration period, the equipment shall be cycled periodically through its phases of operation. Should a failure occur, it should be repaired and tests started over, except the 10 -minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests. Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests, The following equipments shall be selected for Sampling “Tests:

Quantity of equipments Offered for Acceptance	Quantity to be Tested
First 10	1.
11 to 100	2 additional
101 to 200	1 additional
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests after successful completion of the tests in section 4.3.1.

(a) Complete operational tests at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Repeat the above test using the highest and lowest specified input voltage and frequenciess.

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(c) Perform operational test at environmental conditions when the results from the preproduction test indicate that these environmental conditions cause degradations) which result in marginal or unsatisfactory performance.

4.3.3 Special Tests. Special Tests shall be conducted on quantity of equipments for purpose of checking the effect of any design or material change on the performance of equipment and to assure adequate quality con. trol. The equipment selected for special tests may be selected from equipment previously subjected to the sampling tests.

4.3.3.1 Special Test Schedule. Selection of equipments for special tests shall be made as follows:

(a) On an early production equipment.

(b) On an equipment midway through the production run.

(c) On an early equipment after an extensive engineering change.

(d) When failure reports or other information indicate additional equipment should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control. The maximum number of special tests shall be defined by the procuring agency at the time of request for bids.

4.3.3.2 Scope of Tests. Special tests shall consist of tests selected by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted:

(a) Temperature - Altitude Tests

(b) Vibration Tests (90 minutes cycling in each of 3 planes)

(c) Humidity Tests

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4.3.4 Equipment Failure. Should a failure occur during either the sampling tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test. A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in section 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be tested
First 25	1
26 to 200	1 additional
201 to 500	1 additional
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions. The life test shall be conducted under the following environmental conditions:

(a) Temperature	-	Room Ambient
(b) Altitude		Normal Ground
(c) Humidity		Room Ambient

4.4.2 Test Periods. The test may be run continuously or intermittently. Any period of operation shall be sufficient in duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off and put through various phases of operation.

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4.4.3 Performance Check. At approximately 8-hour intervals during the test a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.4 Reconditioning of Life Test Samples. An equipment which has been subject to the life test shall be reconditioned as follows:

(a) On completion of the life tests, the equipment shall be reworked by the contractor by replacing all tubes and “wear” items. The "wear” items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedure. Preproduction tests, acceptance tests, and life tests shall be performed in accordance with a test plan and procedures mutually agree able to the procuring activity and the contractor. The test plan shall be prepared by the contractor. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the procuring activity will furnish the procedures “to be used for testing.

4.6 Presubmission Testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

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4.8 Equipment Changes. After approval of the preproduction model, no changes which affect weight, installation inter changeability, or interchangeability of maintenance parts of assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Porposal (ECP) giving the necessary information concerning the change. This shall apply for either procuring activity recommended change or contractor's proposed change. When required, the contractor shall also prepare Material Change (MC) and Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with MIL-T-23336.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of equipment shall be preserved packages, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering Data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of packaging and packing (see section 5.1).

6.2 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specifications.

(b) Specification - This specification shall have precedence over all applicable subsidiary s pacifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

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(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Notation of Revisions. A bar in the margin denotes a change made in that paragraph from the previous version of the document. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.4 Availability of Documents. When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

6.5 Connectors.

6.5.1 ECP SM-79. Wiring at connectors noted includes changes authorized by ECP SM-79.

6.5.2 ECP SM-46. Wiring at connectors noted includes changes authorized by ECP SM-46.

6. 5.3 ECP SM-101. Wiring at connectors noted includes changes authorized by ECP SM-101.

6.5.4 ECP SM-106. Wiring at connectors noted includes changes authorized by ECP SM-106.

6.5.5 ECP SM-125. Wiring at connectors noted includes changes authorized by ECP SM-125.

6.5.6 ECP SM-83/99. Wiring at connectors noted includes changes authorized by ECP SM(A)-83/99.

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6.5.7 ECP SM-124. Wiring at connectors noted includes changes authorized by ECP SM(A)-124.

6.5.8 ECP SM-128/69R/123, Wiring at connectors noted includes changes authorized by ECP SM-128, SM(A)-69R2, and SM-123.

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81527B(AS) TEST SET, MISSILE CONTROL SYSTEM, AWG-10 SIDE BENCH

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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